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UNIT 8

SUPPLY AND DEMAND: PRICE-TAKING AND COMPETITIVE MARKETS

How markets operate when all buyers and sellers are price-takers

- Competition can constrain buyers and sellers to be price-takers.
- The interaction of supply and demand determines a market equilibrium in which both buyers and sellers are price-takers, called a competitive equilibrium.
- Prices and quantities in competitive equilibrium change in response to supply and demand shocks.
- Price-taking behaviour ensures that all gains from trade in the market are exhausted at a competitive equilibrium.
- The model of perfect competition describes idealized conditions under which all buyers and sellers are price-takers.
- Real-world markets are typically not perfectly competitive, but some policy problems can be analysed using this demand and supply model.
- There are important similarities and differences between price-taking and price-setting firms.

Students of American history learn that the defeat of the southern Confederate states in the American Civil War ended slavery in the production of cotton and other crops in that region. There is also an economics lesson in this story.

At the war's outbreak on 12 April 1861, President Abraham Lincoln ordered the US Navy to blockade the ports of the Confederate states. These states had declared themselves independent of the US to preserve the institution of slavery.

As a result of the naval blockade, the export of US-grown raw cotton to the textile mills of Lancashire in England came to a virtual halt, eliminating three-quarters of the supply of this critical raw material. Sailing at night, a few blockade-running ships evaded Lincoln's patrols, but 1,500 were destroyed or captured.

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We will see in this unit that the market price of a good, such as cotton, is determined by the interaction of supply and demand. In the case of raw cotton, the tiny quantities reaching England through the blockade were a dramatic reduction in supply. There was large **excess demand**—that is to say, at the prevailing price, the quantity of raw cotton demanded exceeded the available supply. As a result, some sellers realized they could profit by raising the price. Eventually, cotton was sold at prices six times higher than before the war, keeping the lucky blockade-runners in business. Consumption of cotton fell to half the prewar level, throwing hundreds of thousands of people who worked in cotton mills out of work.

Mill owners responded. For them, the price rise was an increase in their costs. Some firms failed and left the industry due to the reduction in their profits. Mill owners looked to India to find an alternative to US cotton, greatly increasing the demand for cotton there. The excess demand in the markets for Indian cotton gave some sellers an opportunity to profit by raising prices, resulting in increases in the prices of Indian cotton, which quickly rose almost to match the price of US cotton.

Responding to the higher income now obtainable from growing cotton, Indian farmers abandoned other crops and grew cotton instead. The same occurred wherever cotton could be grown, including Brazil. In Egypt, farmers who rushed to expand the production of cotton in response to the higher prices began employing slaves, captured (like the American slaves that Lincoln was fighting to free) in sub-Saharan Africa.

There was a problem. The only source of cotton that could come close to making up the shortfall from the US was in India. But Indian cotton differed from American cotton, and required an entirely different kind of processing. Within months of the shift to Indian cotton, new machinery was developed to process it.

As the demand for this new equipment soared, firms like Dobson and Barlow, who made textile machinery, saw profits take-off. We know about this firm, because detailed sales records have survived. It responded by increasing production of these new machines and other equipment. No mill could afford to be left behind in the rush to retool, because if it didn't, it could not use the new raw materials. The result was, in the words of Douglas Farnie, a historian who specialized in the history of cotton production, 'such an extensive investment of capital that it amounted almost to the creation of a new industry.'

The lesson for economists: Lincoln ordered the blockade, but in what followed, the farmers and sellers who increased the price of cotton were not responding to orders. Neither were the mill owners who cut back the output of textiles and laid off the mill workers, nor were the mill owners desperately searching for new material. By ordering new machinery, the mill owners set off a boom in investment and new jobs.

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All of these decisions took place over a matter of months, by millions of people, most of whom were total strangers to one another, each seeking to make the best of a totally new economic situation. American cotton was now scarcer, and people responded, from the cotton fields of Maharashtra in India to the Nile delta, to Brazil, and the Lancashire mills.

To understand how the change in the price of cotton transformed the world cotton and textile production system, think about the prices determined by markets as messages. The increase in the price of US cotton shouted: 'find other sources, and find new technologies appropriate for their use.' Similarly, when the price of petrol rises, the message to the car driver is: 'take the train', which is passed on to the railway operator: 'there are profits to be made by running more train services'. When the price of electricity goes up, the firm or the family is being told: 'think about installing photovoltaic cells on the roof.'

In many cases—like the chain of events that began at Lincoln's desk on 12 April 1861—the messages make sense not only for individual firms and families but also for society: if something has become more expensive then it is likely that more people are demanding it, or the cost of producing it has risen, or both. By finding an alternative, the individual is saving money and conserving society's resources. This is because, in some conditions, prices provide an accurate measure of the scarcity of a good or service. [1]

In planned economies, which operated in the Soviet Union and other central and eastern European countries before the 1990s (discussed in Unit 1), messages about how things would be produced are sent deliberately by government experts. They decide what will be produced and at what price it will be sold. The same is true, as we saw in Unit 6, inside large firms like General Motors, where managers (and not prices) determine who does what.

The amazing thing about prices determined by markets is that individuals do not send the messages, but rather the anonymous interaction of sometimes millions of people. And when conditions change—a cheaper way of producing bread, for example—nobody has to change the message ('put bread instead of potatoes on the table tonight'). A price change results from a change in firms' costs. The reduced price of bread says it all.

8.1 Buying and selling: Demand and supply in a competitive market

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In Unit 7 we considered the case of a differentiated product sold market for such a product there is one seller with many buyers. Now we look at markets in which many buyers and sellers interact, and show how the market price is determined by both the preferences of consumers and the costs of suppliers.

For a simple model of a market with many buyers and sellers, think about the potential for trade in second-hand copies of a recommended textbook for a university economics course. Demand for the book comes from students who are about to begin the course, and they will differ in their **willingness to pay (WTP)**. No one will pay more than the price of a new copy in the campus bookshop. Below that, students' WTP may depend on how hard they work, how important they think the book is, and on their available resources for buying books.

Willingness to pay (WTP) is a useful concept for buyers in online auctions, such as eBay. If you want to bid for an item, one way to do so is to set a maximum bid equal to your WTP, which will be kept secret from other bidders. For how to do this on eBay, see their online customer service centre.

eBay will place bids automatically on your behalf until you are the highest bidder, or until your maximum is reached. You will win the auction if, and only if, the highest bid is less than or equal to your WTP.

Figure 8.1 shows the demand curve. As we did in Unit 7, we line up all the consumers in order of willingness to pay, highest first. The first student is willing to pay Rs. 20, the 20th Rs. 10, and so on. For any price, P, the graph tells you how many students would be willing to buy: it is the number whose WTP is at or above P.

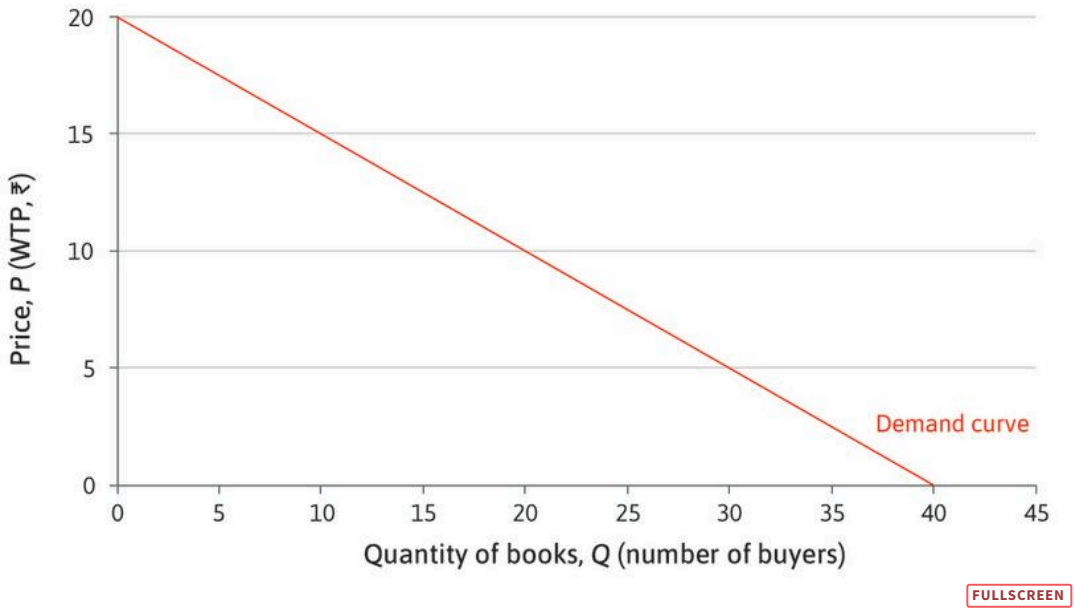


Figure 8.1 The market demand curve for books.

The demand curve represents the WTP of buyers; similarly, supply depends on the sellers' **willingness to accept (WTA)** money in return for books.

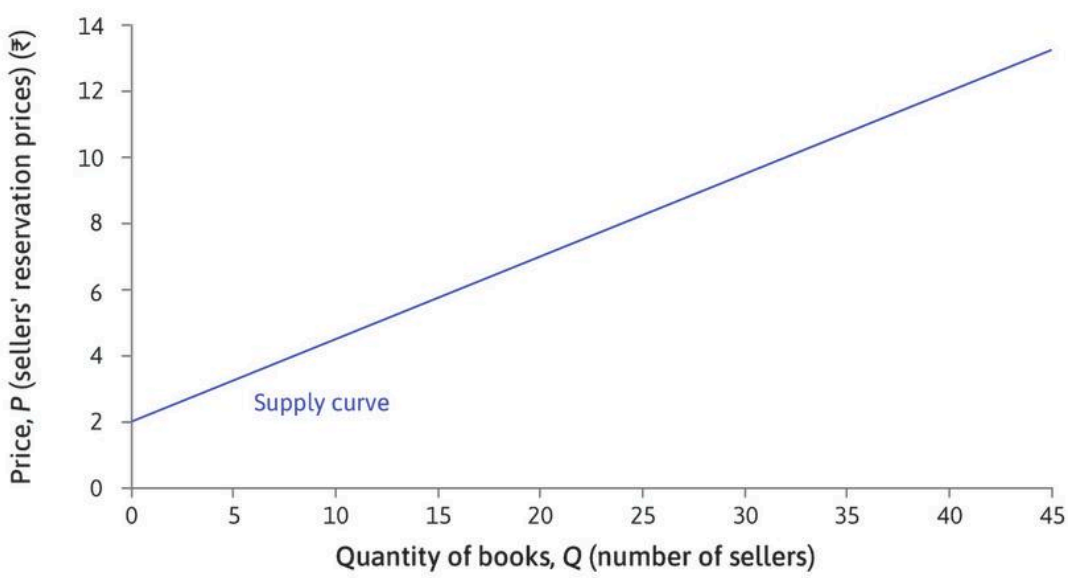
The supply of second-hand books comes from students who hav

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the course, who will differ in the amount they are willing to accept—that is, their **reservation price**. Recall from Unit 5 that Angela was willing to enter into a contract with Bruno only if it gave her at least as much utility as her reservation option (no work and survival rations); here the reservation price of a potential seller represents the value to her of keeping the book, and she will only be willing to sell for a price at least that high. Poorer students (who are keen to sell so that they can afford other books) and those no longer studying economics may have lower reservation prices. Again, online auctions like eBay allow sellers to specify their WTA.

We can draw a **supply curve** by lining up the sellers in order of their reservation prices (their WTAs): see Figure 8.2. We put the sellers who are most willing to sell—those who have the lowest reservation prices—first, so the graph of reservation prices slopes upward.

If you sell an item on eBay you can set a reserve price, which will not be disclosed to the bidders. This article explains eBay reserve prices. You are telling eBay that the item should not be sold unless there is a bid at (or above) that price. So the reserve price should correspond to your WTA. If no one bids your WTA, the item will not be sold.



Reservation price

The first seller has a reservation price of Rs. 2, and will sell at any price above that.

Figure 8.2 The supply curve for books.

For any price, the supply curve shows the number of students willing to sell at that price —that is, the number of books that will be supplied to the market. Notice that we have drawn the supply and demand curves as straight lines for simplicity. In practice they are more likely to be curves, with the exact shape depending on how vary among the students.

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QUESTION 8.1 CHOOSE THE CORRECT ANSWER(S)

LINK

As a student representative, one of your roles is to organize a second-hand textbook market between the current and former first-year students. After a survey, you estimate the demand and supply curves to be the ones shown in Figures 8.1 and 8.2. For example, you estimate that pricing the book at Rs. 7 would lead to a supply of 20 books and a demand of 26 books. Which of the following statements is correct?

- [ ] A rumour that the textbook may be required again in Year 2 would change the supply curve, shifting it upwards.
- [ ] Doubling the price to Rs. 14 would double the supply.
- [ ] A rumour that the textbook may no longer be on the reading list for the first-year students would change the demand curve, shifting it upwards.
- [ ] Demand would double if the price were reduced sufficiently.

Check my answers

EXERCISE 8.1 SELLING STRATEGIES AND RESERVATION PRICES

LINK

Consider three possible methods to sell a car that you own:

- Advertise it in the local newspaper.
- Take it to a car auction.
- Offer it to a second-hand car dealer.

1. Would your reservation price be the same in each case? Why?
2. If you used the first method, would you advertise it at your reservation price?
3. Which method do you think would result in the highest sale price?
4. Which method would you choose?

8.2 The market and the equilibrium price

What would you expect to happen in the market for this textbook? That will depend on the market institutions that bring buyers and sellers together. If students have to rely on

word-of-mouth, then when a buyer finds a seller they can try to negotiate a deal that suits both of them. But each buyer would like to be able to find a seller with a low reservation price, and each seller would like to find a buyer with a high willingness to pay. Before concluding a deal with one trading partner, both par

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about other trading opportunities.

Traditional market institutions often brought many buyers and sellers together in one place. Many of the world's great cities grew up around marketplaces and bazaars along ancient trading routes such as the Silk Road between China and the Mediterranean. In the Grand Bazaar of Istanbul, one of the largest and oldest covered markets in the world, shops selling carpets, gold, leather, and textiles cluster together in different areas. In medieval towns and cities it was common for makers and sellers of a specific type of good to set up shops close to each other, so customers knew where to find them. The city of London is now a financial centre, but evidence of trades once carried out there can be found in surviving street names: Pudding Lane, Bread Street, Milk Street, Threadneedle Street, Ropemaker Street, and Silk Street.

With modern communications, sellers can advertise their goods and buyers can more easily find out what is available, and where to buy it. But in some cases it is still convenient for many buyers and sellers to meet each other. Large cities have markets for meat, fish, vegetables or flowers, where buyers can inspect and compare the quality of the produce. In the past, markets for second-hand goods often involved specialist dealers, but nowadays sellers can contact buyers directly through online marketplaces such as eBay. Websites now help students sell textbooks to others in their university.

At the end of the nineteenth century, the economist Alfred Marshall introduced his model of supply and demand using a similar example to our case of second-hand books. Most English towns had a corn exchange (also known as a grain exchange)—a building where farmers met with merchants to sell their grain. Marshall described how the supply curve of grain would be determined by the prices that farmers would be willing to accept, and the demand curve by the willingness to pay of merchants. Then he argued that, although the price 'may be tossed hither and thither like a shuttlecock' in the 'higgling and bargaining' of the market, it would never be very far from the particular price at which the quantity demanded by merchants was equal to the quantity the farmers would supply.

Marshall called the price that equated supply and demand the *equilibrium price*. If the price was above the equilibrium, farmers would want to sell large quantities of grain. But few merchants would want to buy—there would be **excess supply**. Then, even the merchants who were willing to pay that much would realize that farmers would soon have to lower their prices and would wait until they did. Similarly, if the price was below the equilibrium, sellers would prefer to wait rather than sell at that price. If, at the going price, the amount supplied did not equal the amount demanded, Marshall

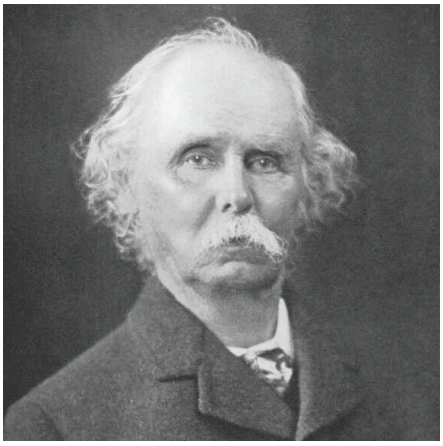
reasoned that some sellers or buyers could benefit by charging some other price (in modern terminology, we would say that the going price was not a **Nash equilibrium**). So the price would tend to settle at an **equilibrium** level, where demand and supply were equated.

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Marshall's argument was based on the assumption that all the grain was of the same quality. His supply and demand model can be applied to markets in which all sellers are selling identical goods, so buyers are equally willing to buy from any seller. If the farmers all had grain of different qualities, they would be more like the sellers of differentiated products in Unit 7.

GREAT ECONOMISTS

Alfred Marshall LINK



Alfred Marshall (1842–1924) was a founder—with Léon Walras—of what is termed the neoclassical school of economics. His *Principles of Economics*, first published in 1890, was the standard introductory textbook for English speaking students for 50 years. An excellent mathematician, Marshall provided new foundations for the analysis of supply and demand by using calculus to formulate the workings of markets and firms, and express key concepts such as **marginal costs** and **marginal utility**. The concepts of consumer and producer surplus are also due to Marshall. His conception of economics as an attempt to 'understand the influences exerted on the quality and tone of a man's life by the manner in which he earns his livelihood …' is close to our own definition of the field. 2

Sadly, much of the wisdom in Marshall's text has rarely been taught by his followers. Marshall paid attention to facts. His observation that large firms could produce at lower unit costs than small firms was integral to his thinking, but it never found a place in the neoclassical school. This may be because if the average cost curve is downward-sloping even when firms are very large, there will be a kind of winner-takes-all competition in which a few large firms emerge as winners with the power to set prices, rather than taking the going price as a given. We return to this problem in Unit 12 and it is explained further in Unit 21 of *The Economy*.

Marshall would also have been distressed that *homo economicus* (whose existence we questioned in Unit 4) became the main actor in textbooks written by the followers of the neoclassical school. He insisted that:

> Ethical forces are among those of which the economist has to take account. Attempts have indeed been made to construct an abstract science with regard to the actions of an economic man who is under no ethical influences and who pursues pecuniary gain ... selfishly. But they have not b
> (*Principles of Economics*, 1890)

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While advancing the use of mathematics in economics, he also cautioned against its misuse. In a letter to A. L. Bowley, a fellow mathematically inclined economist, he explained his own 'rules' as follows:

1. Use mathematics as a shorthand language, rather than as an engine of inquiry
2. Keep to them [that is, stick to the maths] till you have done
3. Translate into English
4. Then illustrate by examples that are important in real life
5. Burn the mathematics
6. If you can't succeed in 4, burn 3: 'This I do often.'

Marshall was Professor of Political Economy at the University of Cambridge between 1885 and 1908. In 1896 he circulated a pamphlet to the University Senate objecting to a proposal to allow women to be granted degrees. Marshall prevailed and women would wait until 1948 before being granted academic standing at Cambridge on a par with men.

But his work was motivated by a desire to improve the material conditions of working people:

> Now at last we are setting ourselves seriously to inquire whether it is necessary that there should be any so called lower classes at all: that is whether there need be large numbers of people doomed from their birth to hard work in order to provide for others the requisites of a refined and cultured life, while they themselves are prevented by their poverty and toil from having any share or part in that life. ... The answer depends in a great measure upon facts and inferences, which are within the province of economics; and this is it which gives to economic studies their chief and their highest interest. (*Principles of Economics*, 1890)

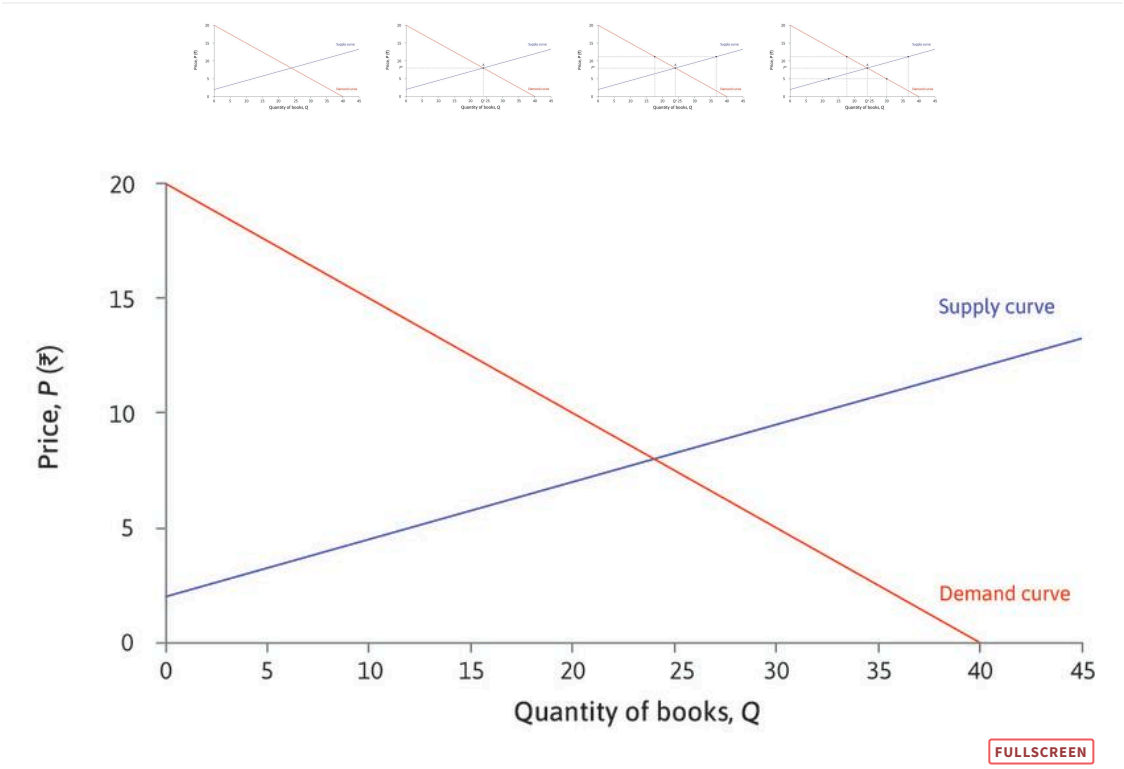
Would Marshall now be satisfied with the contribution that modern economics has made to creating a more just economy?

To apply the supply and demand model to the textbook market, we assume that all the books are identical (although in practice some may be in better condition than others) and that a potential seller can advertise a book for sale by announcing its price on a local website. As at the Corn Exchange, we would expect that most trades would occur at

similar prices. Buyers and sellers can easily observe all the advertised prices, so if some books were advertised at Rs. 10 and others at Rs. 5, buyers would be queuing to pay Rs. 5. These sellers would quickly realize that they could charge more, while no one would want to pay Rs. 10 so these sellers would have to lower their pric

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We can find the equilibrium price by drawing the supply and demand curves on one diagram, as in Figure 8.3. At a price P^* = Rs. 8, the supply of books is equal to demand: 24 buyers are willing to pay Rs. 8, and 24 sellers are willing to sell. The equilibrium quantity is Q^* = 24.



Supply and demand

We find the equilibrium by drawing the supply and demand curves in the same diagram.

Figure 8.3 Equilibrium in the market for second-hand books.

The **market-clearing price** is Rs. 8—that is, supply is equal to demand at this price, so all buyers who want to buy and all sellers who want to sell can do so. The market is in equilibrium. In everyday language, something is in equilibrium if the forces acting on it are in balance, so that it remains still. Remember Fisher's hydraulic model of price determination from Unit 2: changes in the economy caused water to flow through the apparatus until it reached an equilibrium, with no further tendency for prices to change. We say that a market is in **equilibrium** if the actions of buyers and sellers have no tendency to change the price or the quantities bought and sold, unless there is a change in market conditions such as the numbers of potential buyers and sellers, and how

much they value the good. At the equilibrium price for textbooks, all those who wish to buy or sell are able to do so, so there is no tendency for change.

Price-taking LINK

Will the market always be in equilibrium? As we have seen, Marshall argued that prices would not deviate far from the equilibrium level, because people would want to change their prices if there were excess supply or demand. In this unit, we study competitive market equilibria. In Unit 11 we will look at when and how prices change when the market is not in equilibrium.

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Not all online markets for books are in competitive equilibrium. In one case when the conditions for equilibrium were not met, automatic price-setting algorithms raised the price of a book to $23 million! Michael Eisen, a biologist, noticed a classic but out-of-print text, *The Making of a Fly*, was listed for sale on Amazon by two reputable sellers, with prices starting at $1,730,045.91 (+$3.99 shipping). He watched over the next week as the prices rose rapidly, eventually peaking at $23,698,655.93, before dropping to $106.23. Eisen explains why in his blog.

In the textbook market that we have described, individual students have to accept the prevailing equilibrium price in the market, determined by the supply and demand curves. No one would trade with a student asking a higher price or offering a lower one, because anyone could find an alternative seller or buyer with a better price. The participants in this market are **price-takers**, because there is sufficient competition from other buyers and sellers so the best they can do is to trade at the same price. Any buyer or seller is of course free to choose a different price, but they cannot benefit by doing so.

We have seen examples where market participants do not behave as price-takers: the producer of a differentiated product can set its own price because it has no close competitors. Notice, however, that although the *sellers* of differentiated products are price-setters, the *buyers* in Unit 7 were price-takers. Since there are so many consumers wanting to buy language courses, an individual consumer has no power to negotiate a more advantageous deal, but simply has to accept the price that all other consumers are paying.

In this unit, we study market equilibria where both buyers and sellers are price-takers. We expect to see price-taking on both sides of the market where there are many sellers selling the identical goods, and many buyers wishing to purchase them. Sellers are forced to be price-takers by the presence of other sellers, as well as buyers who always choose the seller with the lowest price. If a seller tried to set a higher price, buyers would simply go elsewhere.

Similarly buyers are price-takers when there are plenty of other buyers, and sellers willing to sell to whoever will pay the highest price. On both sides of the market, competition eliminates bargaining power. We will describe the equilibrium in such a market as a **competitive equilibrium**.

A competitive market equilibrium is a Nash equilibrium, because given what all other actors are doing (trading at the equilibrium price), no actor can do better than to continue what he or she is doing (also trading at the equilibrium price).

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EXERCISE 8.2 PRICE-TAKERS LINK

Think about some of the goods you buy: perhaps different kinds of food, clothes, transport tickets, or electronic goods.

1. Are there many sellers of these goods?
2. Do you try to find the lowest price in each case?
3. If not, why not?
4. For which goods would price be your main criterion?
5. Use your answers to help you decide whether the sellers of these goods are price-takers. Are there goods for which you, as a buyer, are not a price-taker?

QUESTION 8.2 CHOOSE THE CORRECT ANSWER(S)

LINK

The diagram shows the demand and the supply curves for a textboo

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at $(Q, P) = (24, 8)$. Which of the following is correct?

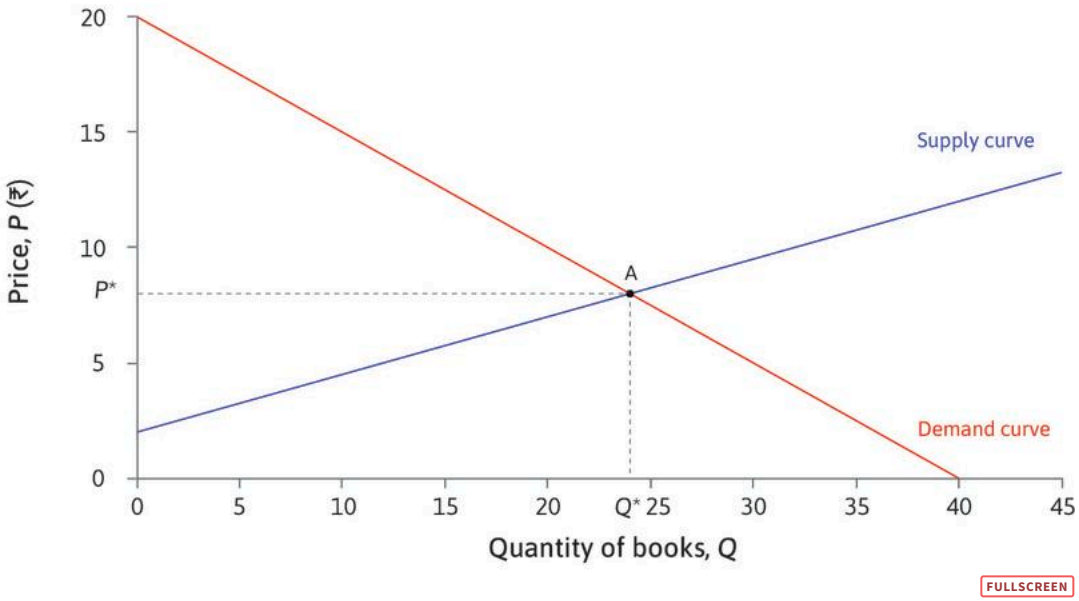


Figure 8.4 Supply and demand in the market for textbooks.

- [ ] At price Rs. 10, there is an excess demand for the textbook.
- [ ] At Rs. 8, some of the sellers have an incentive to increase their selling price to Rs. 9.
- [ ] At Rs. 8, the market clears.
- [ ] 40 books will be sold in total.

Check my answers

8.3 Demand and supply in a competitive market: Bakeries

In the second-hand textbook example, both buyers and sellers are individual consumers. Now we look at markets where the sellers are firms. We know from Unit 7 how firms choose their price and quantity when producing differentiated goods, and we saw that if other firms made similar products, their choice of price would be restricted (the demand curve for their own product would be almost flat) as raising the price would cause consumers to switch to other similar brands.

If there are many firms producing identical products, and consumers can easily switch from one firm to another, then firms will be price-takers in equilibrium. They will be unable to benefit from attempting to trade at a price different from the prevailing price.

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To understand how price-taking firms behave, consider a city in which many small bakeries produce bread and sell it directly to consumers. Figure 8.5 shows what the market demand curve (the total daily demand for bread of all consumers in the city) might look like. It is downward-sloping as usual because, at higher prices, fewer consumers will be willing to buy.

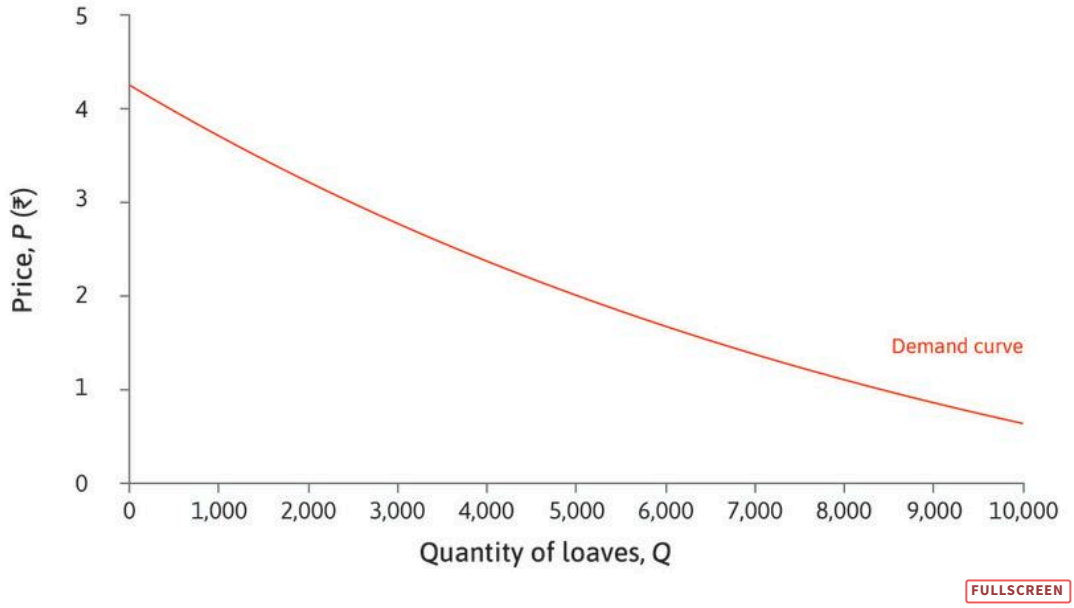


Figure 8.5 The market demand curve for bread.

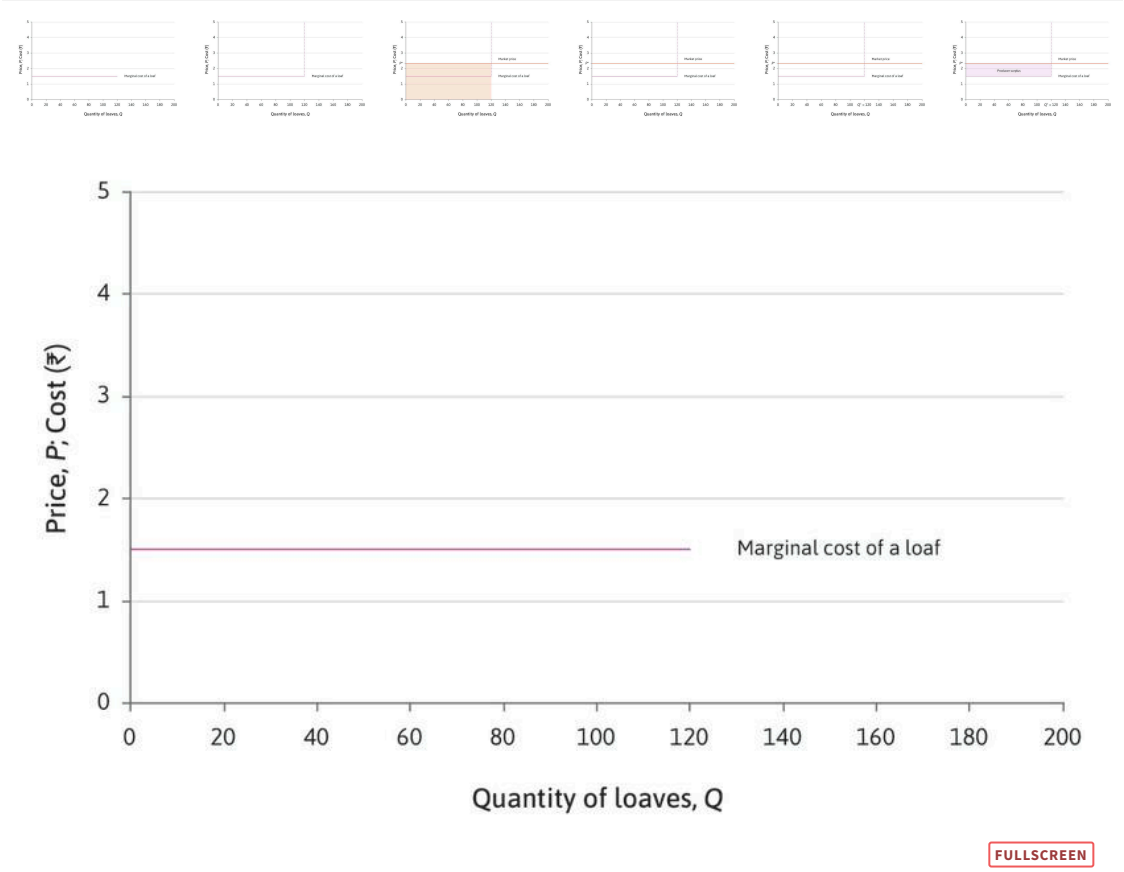
Suppose that you are the owner of one small bakery. You have to decide what price to charge and how many loaves to produce each morning. Suppose that neighbouring bakeries are selling loaves identical to yours at Rs. 2.35. The loaf is a large baguette. This is the prevailing market price; you will not be able to sell loaves at a higher price than other bakeries, because no one would buy—you are a price-taker.

What you should do depends on your costs of production—and, in particular, on your marginal costs. You may have some fixed costs, for example, the rent you pay—but you have to pay these irrespective of the number of loaves you make. It is the additional costs of actually making each loaf of bread—the cost of the ingredients, and what you have to pay your employees for the time it takes to bake a loaf—that determine whether you should produce 30, 50 or 100 loaves per day. Having installed mixers, ovens, and other equipment, the marginal cost of each extra loaf may be relatively low, as long as you don't exceed the capacity of your equipment.

Figure 8.6 illustrates this situation. Your bakery has the capacity to produce up to 120 loaves per day. Below that level, the marginal cost of a loaf is Rs. 1.50. The horizontal line at P^* = Rs. 2.35 represents the demand for bread from your bakery—because you are a price-taker, each loaf you produce can be sold for Rs. 2.35.

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In this example, it is easy to find the profit-maximizing price and quantity without drawing isoprofit curves. Work through the analysis of Figure 8.6 to see how this is done.



The marginal cost of a loaf

Whatever quantity you decide to produce between 0 and 120, the cost of making one more loaf, that is, the marginal cost, is Rs. 1.50.

Figure 8.6 The profit-maximizing price and quantity.

Your optimal choice is P^* = Rs. 2.35 and Q = 120; since your marginal cost is less than the market price, you maximize profit by making as many loaves as possible. Your profit will be the total surplus on 120 loaves minus your fixed costs.

What would you do if the market price changed? As long as it remains above Rs. 1.50, your profit-maximizing choice remains the same. But if the price fell below Rs. 1.50, you

should immediately stop making bread. In that case, you would make a loss on any loaf produced.

Even if the market price were a little above Rs. 1.50, you might m Continue from your last visit? fixed costs were high. In this case, you still do the best you can by producing 120 loaves —at least the surplus will help you to cover part of the fixed costs. But in the longer term you would need to consider whether it is worth continuing in business. If you expect market conditions to remain bad, it might be best to sell up and leave the market —you could obtain a better return on your capital elsewhere.

The market supply curve LINK

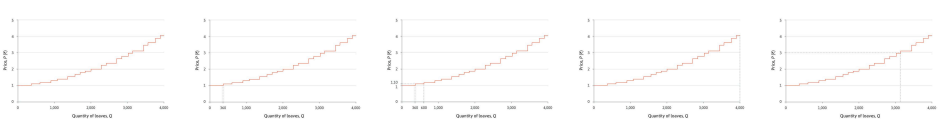
The market for bread in the city has many consumers and many bakeries. Let's suppose initially there are 20 bakeries, differing in their marginal costs and their production capacity. Costs differ across bakeries because they specialize in producing different kinds of bread. Those that specialize in making large baguettes have the lowest marginal costs; for other bakeries to supply this type of bread, they need to switch employees who are more skilled in other types of bread production to baguette baking and their costs are higher. Similarly, the equipment of the non-specialist bakeries is less suited to producing baguettes, which is another reason their marginal costs are higher. As bakeries less and less suited to produced large baguettes supply the market, marginal costs rise, as shown in Figure 8.7.

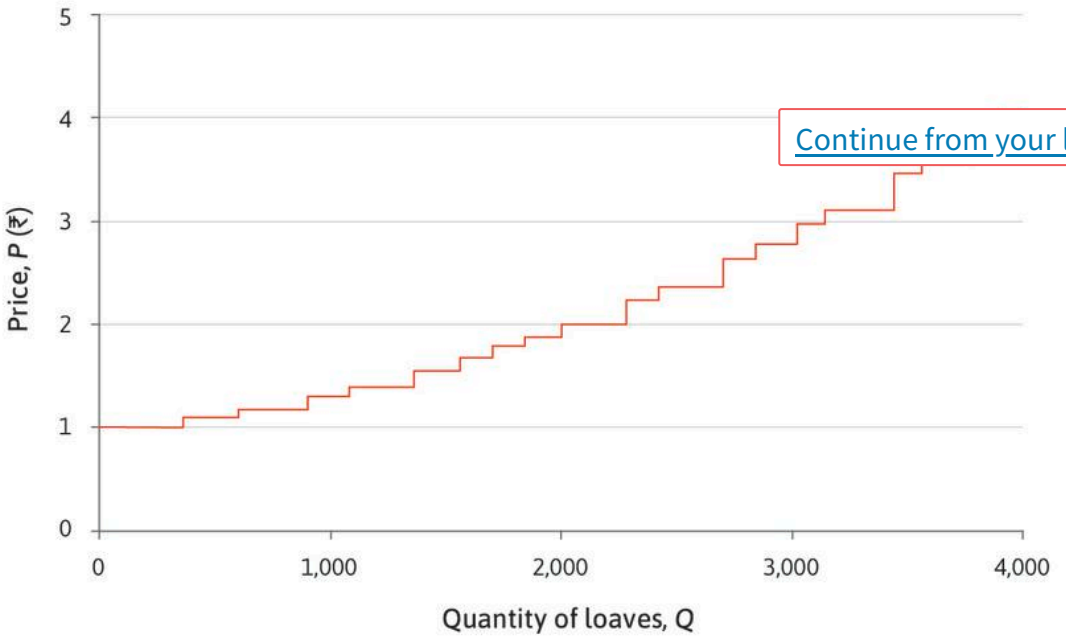
Each bakery will decide how many loaves to produce in the same way:

- *When the market price is above its marginal cost*: It will produce and sell its maximum output.
- *When the market price is below its marginal cost*: It will make none of this kind of bread.

We can work out how much each bakery will supply at any given market price. To find the market supply curve, we just add up the total amount that all the bakeries will supply at each price.

We can do this in the same way as for second-hand textbooks. Figure 8.7 shows how to find the market supply by lining the 20 bakeries up in order of their marginal costs.





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The market supply curve

To draw the market supply, we line up the 20 bakeries in order of their marginal costs—lowest first—and plot the marginal cost of each one, up to the maximum number of loaves it can produce.

Figure 8.7 The market supply curve: 20 bakeries.

If there were many more bakeries in the city, more bread would be produced and there would be many more 'steps' on the supply curve. Rather than drawing them all, it is easier to approximate market supply with a smooth curve. Figure 8.8 shows an approximate market supply curve when there are many firms.

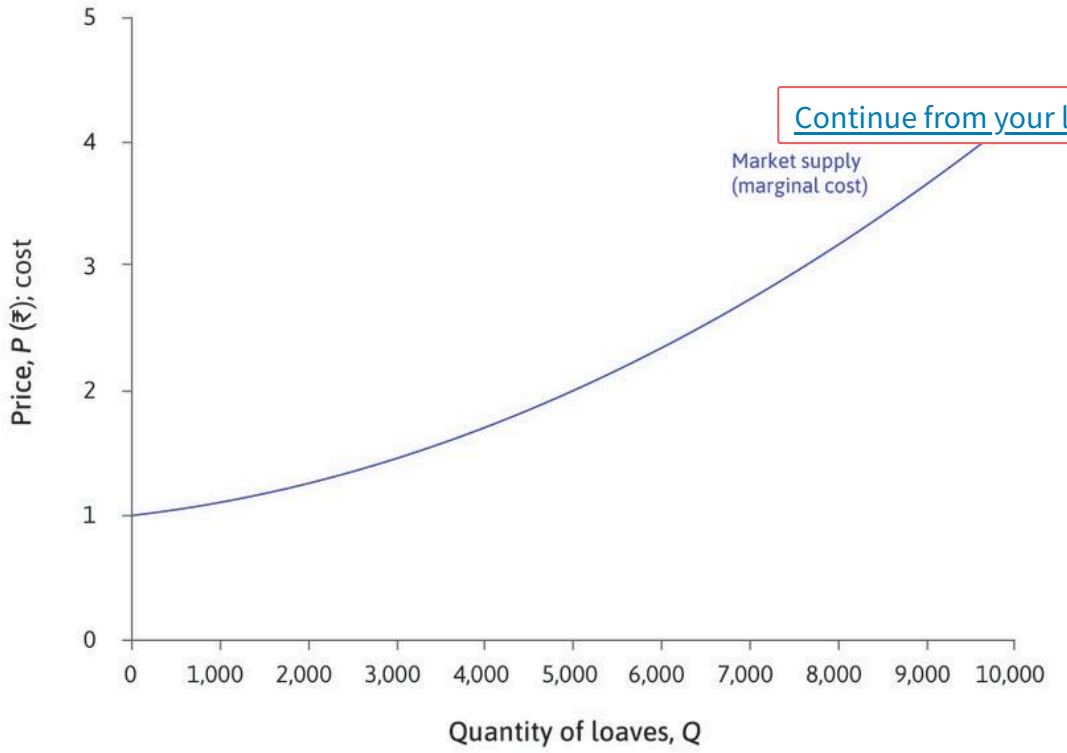
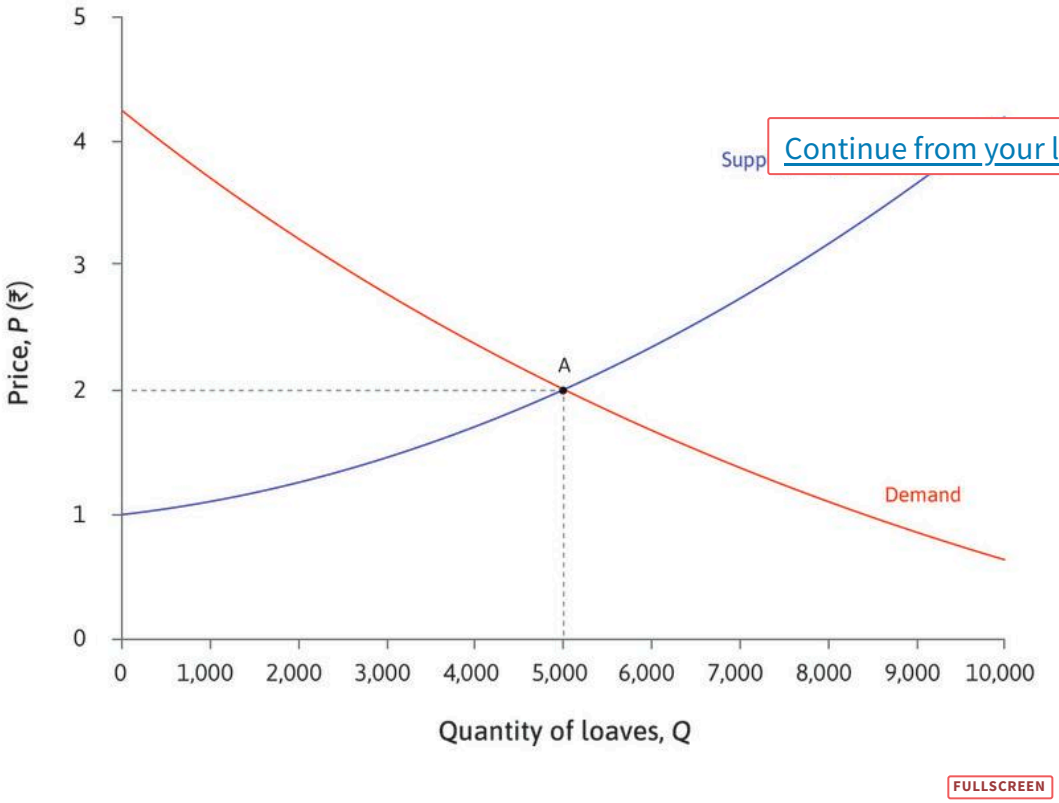


Figure 8.8 The market supply curve: Many bakeries.

Notice that the supply curve tells us two different things. If we choose any price, it tells us how many loaves, in total, the bakeries would produce. But remember that, to construct it, we plotted the marginal cost of each loaf of bread in increasing order of marginal costs. So, if we choose a particular quantity (7,000, say) and use the curve to find the corresponding value on the vertical axis (Rs. 2.74), this tells us that the marginal cost of the 7,000th loaf is Rs. 2.74. In other words, the market supply curve is the marginal cost curve for all the bread produced in the city.

Competitive equilibrium LINK

Now we know both the demand curve (Figure 8.5) and the supply curve (Figure 8.8) for the bread market as a whole. Figure 8.9 shows that the competitive equilibrium price is exactly Rs. 2.00. At this price, the market clears—consumers demand 5,000 kilograms per day, and firms supply 5,000 loaves per day.



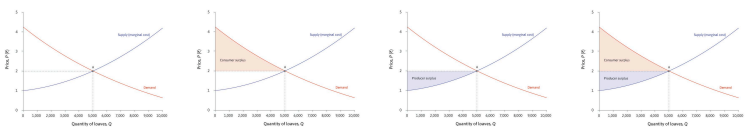
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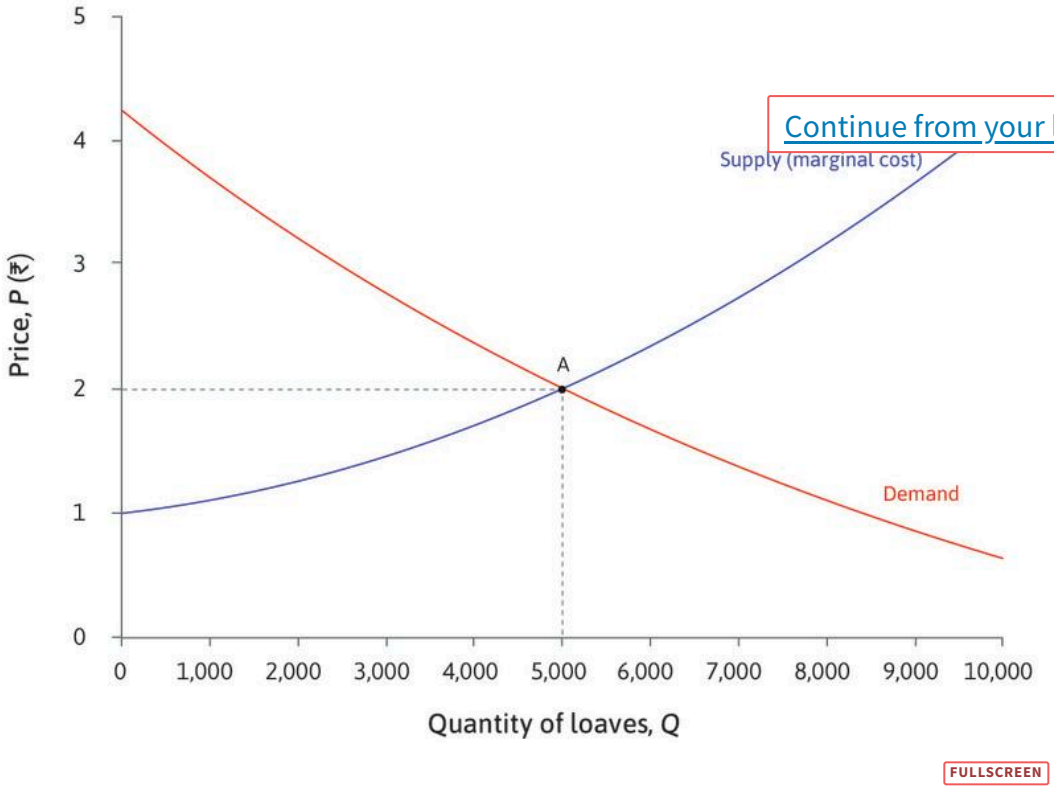
Figure 8.9 Equilibrium in the market for bread.

Since the equilibrium is the point where the demand curve crosses the marginal cost curve, we know that—in equilibrium—both the willingness to pay of the 5,000th consumer, and the marginal cost of the 5,000th loaf, are equal to the market price.

8.4 Competitive equilibrium: Gains from trade, allocation, and distribution

Buyers and sellers of bread voluntarily engage in trade because both benefit. Their mutual benefits from the equilibrium allocation can be measured by the consumer and producer surpluses introduced in Unit 7. Any buyer whose willingness to pay for a good is higher than the market price, receives a surplus—the difference between the WTP and the price paid. Similarly, if the marginal cost of producing an item is below the market price, the producer receives a surplus. Figure 8.10 shows how to calculate the total surplus (the gains from trade) at the competitive equilibrium in the market for bread, in the same way as we did for the market for language courses.





The consumer surplus

At the equilibrium price of Rs. 2 in the bread market, a consumer who is willing to pay Rs. 3.50 obtains a surplus of Rs. 1.50.

Figure 8.10 Equilibrium in the bread market: Gains from trade.

When the market for bread is in equilibrium with the quantity of loaves supplied equal to the quantity demanded, the total surplus is the area below the demand curve and above the supply curve.

Notice how the equilibrium allocation in this market differs from the allocation of a differentiated product, such as LP's language course. The equilibrium quantity of bread is at the point where the market supply curve (which is also the marginal cost curve) crosses the demand curve; the total surplus is the whole of the area between the two curves.

The competitive equilibrium allocation of bread has the property that the total surplus is maximized. The surplus would be smaller if fewer than 5,000 loaves were produced; if more than 5,000 loaves were produced, the surplus on the extra loaves would be negative—they would cost more to make than consumers were willing to pay.

One may ask which kinds of firms can be expected to behave as price-takers, like the hypothetical example of bakeries in this model. Some firms do operate more or less

under conditions in which they have little control over prices and in which their costs and capacities determine how much they produce. In developing economies like India, small scale agriculture or small vendors are examples of this kind of firm.

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At the equilibrium, all the potential gains from trade are exploited, which means there is no deadweight loss. This property—that the combined consumer and producer surplus is maximized at the point where supply equals demand—holds in general. If both buyers and sellers are price-takers, the competitive equilibrium allocation maximizes the sum of the gains achieved by trading in the market.

Joel Waldfogel, an economist, gave his chosen discipline a bad name by suggesting that gift-giving at Christmas may result in a deadweight loss. If you receive a gift that is worth less to you than it cost the giver, he argued that the surplus from the transaction is negative. Do you agree?

Pareto efficiency LINK

At the competitive equilibrium allocation of bread, it is not possible to make any of the consumers or firms better off (that is, to increase the surplus of any individual) without making at least one of them worse off. Provided that what happens in this market does not affect anyone other than the participating buyers and sellers, we can say that the equilibrium allocation is Pareto efficient.

Pareto efficiency follows from three assumptions we have made about the bread market.

Price-taking

The participants are price-takers. They have no market power. When a particular buyer trades with a particular seller, each of them knows that the other can find an alternative trading partner willing to trade at the market price. Competition prevents sellers from raising the price in the way that the producer of a differentiated good would do.

A complete contract

The exchange of a loaf of bread for money is governed by a complete contract between buyer and seller. If you find there is no loaf of bread in the bag marked ‘Bread’ when you get home, you can get your money back. Compare this with the incomplete employment contract in Unit 6, in which the firm can buy the worker’s time, but cannot be sure how much effort the worker will put in. We will see in Unit 9 that this leads to a Pareto-inefficient allocation in the labour market.

No effects on others

We have implicitly assumed that what happens in this market affects no one except the buyers and sellers. To assess Pareto efficiency, we need to consider everyone affected by the allocation. If, for example, the early morning activities of bakeries disrupt the sleep

of local residents, then there are additional costs of bread production and we ought to take the costs to the bakeries' neighbours into account too. Then, we may conclude that the equilibrium allocation is not Pareto efficient after all. We will investigate this type of problem in Unit 11.

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Fairness LINK

Remember from Unit 5 that there are two criteria for assessing an allocation: efficiency and fairness. Even if we think that the market allocation is Pareto efficient, we should not conclude that it is necessarily a desirable one. What can we say about fairness in the case of the bread market? We could examine the distribution of the gains from trade between producers and consumers. Figure 8.10 shows that both consumers and firms obtain a surplus; in this example consumer surplus is slightly higher than producer surplus. You can see that this happens because the demand curve is relatively steep (inelastic) compared with the supply curve.

We might also want to consider the market participants' standard of living. For example, if a poor student buys a book from a rich student, we might think that an outcome in which the buyer paid less than the market price (closer to the seller's reservation price) would be better, because it would be fairer. Or, if the consumers in the bread market were exceptionally poor, we might decide to pass a law setting a maximum bread price lower than Rs. 2.00 to achieve a fairer (although Pareto-inefficient) outcome.

The Pareto efficiency of a competitive equilibrium allocation is often interpreted as a powerful argument in favour of markets as a means of allocating resources. But we need to be careful not to exaggerate the value of this result:

Maurice Stucke, an antitrust lawyer, asks if competition in a market economy is always good.

- *The allocation may not be Pareto efficient*: We might not have taken everything into account.
- *There are other important considerations*: Fairness, for example.
- *Price-takers are hard to find in real life*: It is not as easy as you might think to find markets where all participants are price-takers. Goods (including bread) are rarely identical, and participants don't always know what prices are available.

Watch economist Kathryn Graddy explain how she collected data on the price of whiting from the Fulton Fish Market in New York and what she found out about the model of perfect competition.

Kathryn Graddy: Fishing for perfect competition
CORE Econ

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Watch on

Kathryn Graddy: Fishing for perfect competition

EXERCISE 8.3 SURPLUS AND DEADWEIGHT LOSS LINK

1. Sketch a diagram to illustrate the competitive market for bread, showing the equilibrium where 5,000 loaves are sold at a price of Rs. 2.00.
2. Suppose that the bakeries get together to form a cartel. They agree to raise the price to Rs. 2.70, and jointly cut production to supply the number of loaves that consumers demand at that price. Shade the areas on your diagram to show the consumer surplus, producer surplus, and deadweight loss caused by the cartel.
3. For what kinds of goods would you expect the supply curve to be highly elastic?
4. Draw diagrams to illustrate how the share of the gains from trade obtained by producers depends on the elasticity of the supply curve.

QUESTION 8.3 CHOOSE THE CORRECT ANSWER(S)

LINK

Which of the following statements about a competitive equilibrium allocation are correct?

- [ ] It is the best possible allocation.
- [ ] No buyer's surplus or seller's surplus can be increased without reducing someone else's surplus.
- [ ] The allocation must be Pareto efficient.
- [ ] The total surplus from trade is maximized.

Check my answers

QUESTION 8.4 CHOOSE THE CORRECT ANSWER(S)

LINK

Figure 8.10 shows the demand and supply curves in the bread mark
of surplus in competitive equilibrium. *Ceteris paribus*, which of the following would affect the distribution of gains from trade between consumers and producers?

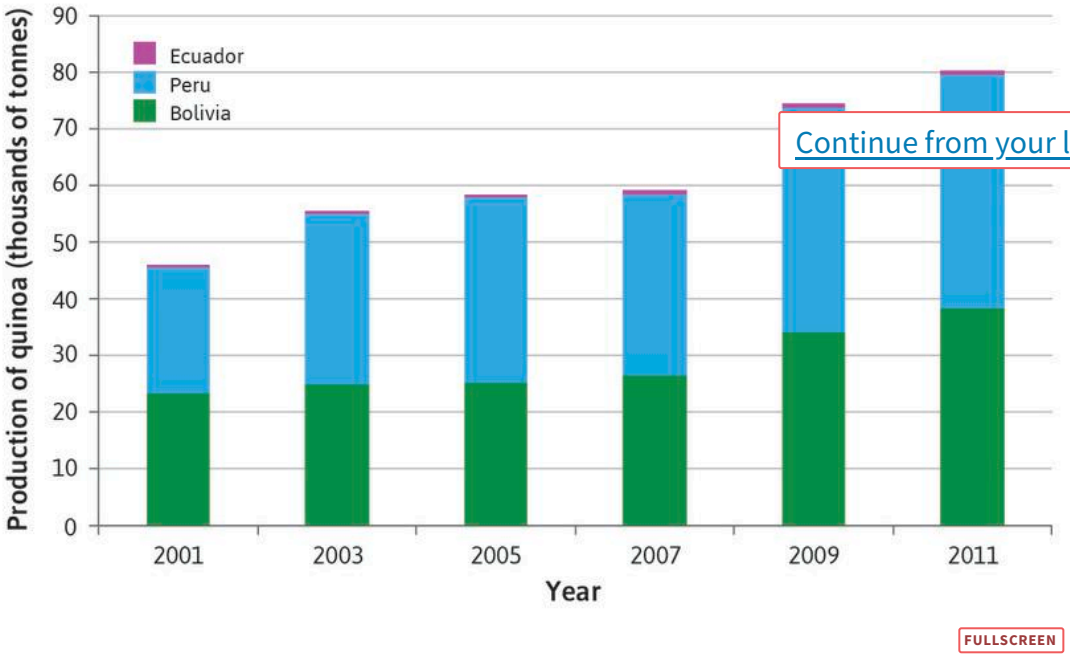
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- [ ] Legislation that sets the price above or below P^*.
- [ ] Relative elasticities of demand and supply.
- [ ] Total quantity traded.
- [ ] The slope of the firms' marginal cost curve.

Check my answers

8.5 Changes in supply and demand

Quinoa is a cereal crop grown on the Altiplano, a high barren plateau in the Andes of South America. It is a traditional staple food in Peru and Bolivia. In recent years, as its nutritional properties have become known, there has been a huge increase in demand from richer, health-conscious consumers in Europe and North America. Figures 8.11a–c show how the market changed. You can see in Figures 8.11a and 8.11b that, between 2001 and 2011, the price of quinoa trebled and production almost doubled. Figure 8.11c indicates the strength of the increase in demand, in real terms, spending on imports of quinoa rose from just $2.4 million to $43.7 million in 10 years. Note that the increase in spending on quinoa reflects the increase in the price shown in Figure 8.11b as well as the purchase of higher quantities of the grain.

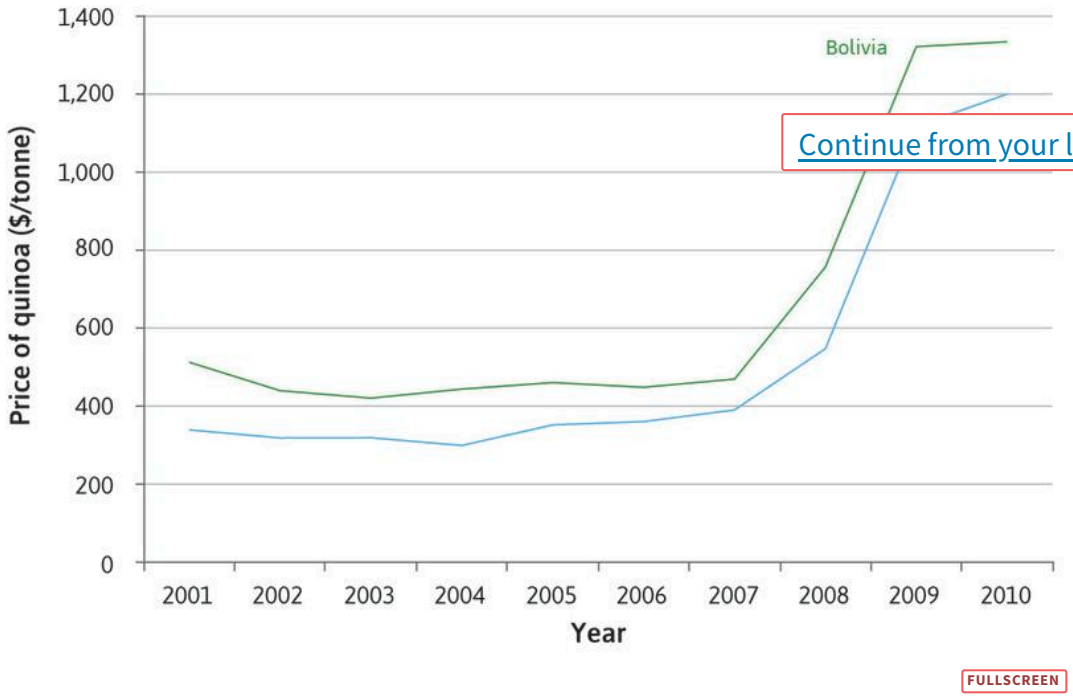


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Figure 8.11a The production of quinoa.

Jose Daniel Reyes and Julia Oliver. 'Quinoa: The Little Cereal That Could'. *The Trade Post*. 22 November 2013. Underlying data from Food and Agriculture Organization of the United Nations. FAOSTAT Database.

For the producer countries these changes are a mixed blessing. While their staple food has become expensive for poor consumers, farmers—who are amongst the poorest—are benefiting from the boom in export sales. Other countries are now investigating whether quinoa can be grown in different climates, and France and the US have become substantial producers.



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Figure 8.11b Quinoa real producer prices in Peru.

Jose Daniel Reyes and Julia Oliver. 'Quinoa: The Little Cereal That Could'. *The Trade Post*. 22 November 2013. Underlying data from Food and Agriculture Organization of the United Nations. FAOSTAT Database.

How can we explain the rapid increase in the price of quinoa? In this section, we look at the effects of changes in demand and supply, illustrating our model in the real-world case of quinoa.

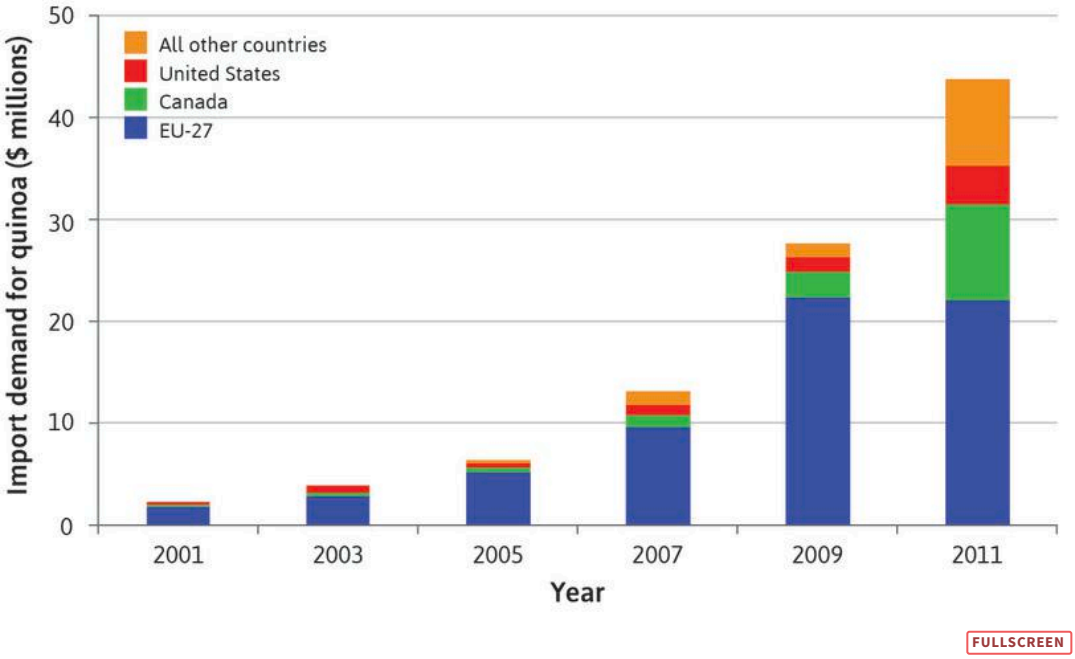


Figure 8.11c Global import demand for quinoa.

Jose Daniel Reyes and Julia Oliver. 'Quinoa: The Little Cereal That Could'. *The Trade Post*. 22 November 2013. Underlying data from Food and Agriculture Organization of the United Nations. FAOSTAT Database.

The supply of quinoa LINK

At the beginning of the current century, there was ample land for growing quinoa on the Altiplano, and farmers producing other crops could easily switch to quinoa as the price rose. As a result, the initial increase in production between 2001 rising demand did not raise costs. This means that the supply curve was virtually flat. But in order to allow the continued output of quinoa after 2008, land less suited for the crop had to be brought into use, and the new farmers taking up the crop were giving up the production of other crops on which they had been making adequate incomes. As a result, costs rose. To represent this, we show the supply curve in Figure 8.13 becoming increasingly steeper as production rose.

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An increase in demand LINK

As in the case of demand for language courses or Apple Cinnamon Cheerios, the demand curve for quinoa sloped downwards, as is shown by 'Original demand (2001)' in Figure 8.13. The original equilibrium was at point A.

The new fashion among North American and European consumers for eating quinoa meant that for any given price of the crop, the tonnes of quinoa purchased rose. In other words, the demand curve for quinoa shifted to the right. You could also say that it shifted up, meaning that the price that was sufficient to allow the sales of any given quantity of quinoa had now increased. This is shown in Figure 8.13 (see the new demand curve labelled '2008').

The increase in demand destroys the old equilibrium (the supply and demand curves no longer cross at A). With the new demand curve, and initially with no change in sales of quinoa or in the price, there were a great many potential consumers whose willingness to pay for quinoa exceeded the price.

Market disequilibrium and adjustment to a new equilibrium

LINK

Thus, point A in Figure 8.13 was no longer a Nash equilibrium because at least some of the producers would have realized that they could raise their prices without reducing their sales. The original price and quantity are termed a **disequilibrium** because, at point A, someone can benefit by changing the price. The reason is that at the initial price, there is excess demand. The increase in demand set off the sequence of events shown in Figure 8.12.

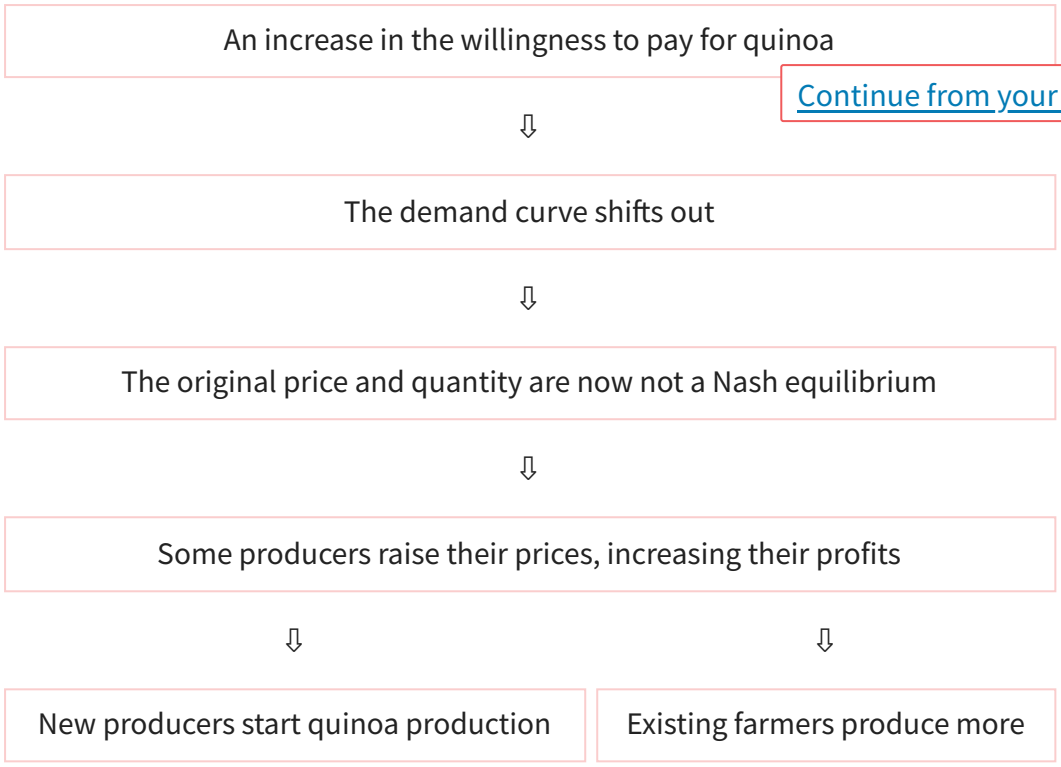
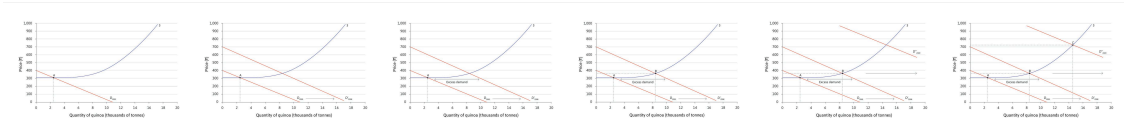
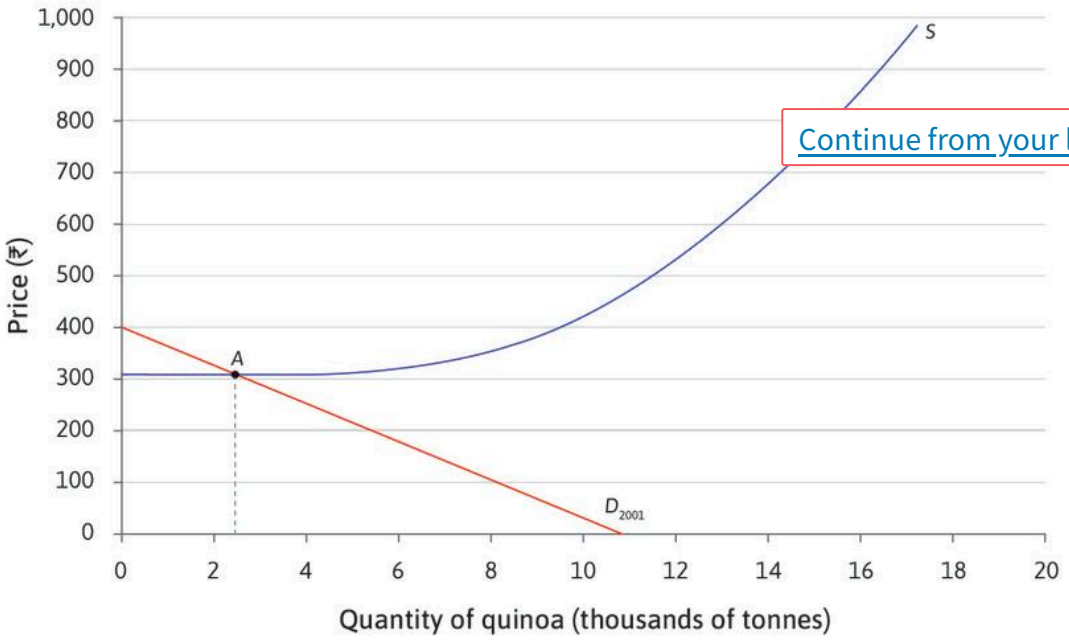


Figure 8.12 Disequilibrium in demand for quinoa.

Figure 8.12 describes what economists call a disequilibrium adjustment process, which is simply how market actors react when the existing situation is not an equilibrium. How long will this process go on? It will continue until the combination of higher prices and greater production has eliminated the excess demand, shown in Figure 8.13 as point C.





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The initial equilibrium point

At the original levels of demand and supply, the equilibrium is at point A. The price is Rs. 340, and Rs. 2.4 million tonnes of quinoa are sold.

Figure 8.13 An increase in the demand for quinoa.

When we refer to 'increase in demand', it's important to be careful about what exactly we mean. Figure 8.14 uses the example of quinoa to explain.

Demand is higher at each possible price (the demand curve has shifted)

⇩

There is a change in the price

⇩

There is an increase in the quantity supplied

Figure 8.14 An increase in demand for quinoa.

This change is a movement along the supply curve. But the supply curve has not shifted! The amount supplied increased in response to the change in price. A shift in the supply curve would take place if, for example, an improved method of cultivating quinoa was invented so that for any given price, more would be supplied.

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After an increase in demand, the equilibrium quantity rises. The price change depends on the steepness of the supply curve, which captures how responsive supply is to prices. We know that the price elasticity of demand measures how much demand falls when prices rise. Similarly, the **price elasticity of supply** is defined as the percentage increase in supply when prices rise by one percent.

You can see in Figure 8.13 that the steeper (more inelastic) the supply curve, the higher the price will rise and the less the quantity will increase. Initially, the supply curve was flat (elastic), so in the equilibrium at point B, the price was the same as at point A and the quantity sold was fully responsive to the demand **shock**.

When either the supply curve or the demand curve shifts, an adjustment of prices is needed to bring the market into equilibrium. Such shifts in supply and demand are often referred to as shocks in economic analysis. We start by specifying an economic model and find the equilibrium. Then we look at how the equilibrium changes when something changes—the model receives a shock. The shock is called **exogenous** because the model doesn't explain why it happened—it shows the consequences of the shock, not the causes.

In the next section, we will examine another example in the world market for oil. Both the supply of and the demand for oil are more elastic in the long run, because producers can eventually build new oil wells and consumers can switch to different fuels for cars or heating. What we mean by the short run in this case is the period during which firms are limited by the capacity of existing wells, and consumers are limited by the cars and heating appliances they currently own.

EXERCISE 8.4 THE MARKET FOR QUINOA LINK

Consider again the market for quinoa studied in Figures 8.11a–c.

1. Would you expect the price to fall eventually to its original level?
2. Use the same model to show the effects on price and quantity of a significant improvement in the methods for producing quinoa, resulting in lower costs for farmers.

EXERCISE 8.5 PRICES, SHOCKS, AND REVOLUTIONS LINK

Historians usually attribute the wave of revolutions in Europe in 1848 to long-term socioeconomic factors and a surge of radical ideas. But a poor wheat harvest in 1845 lead

to food shortages and sharp price rises, which may have contributed to these sudden changes. [3]

Figure 8.15 shows the average and peak prices of wheat from 1838 t
There are three groups of countries: those in which violent revolutio
which constitutional change took place without widespread violence; and those in which no revolution occurred.

Continue from your last visit?

1. Explain, using supply and demand curves, how a poor wheat harvest could lead to price rises and food shortages.
2. Using Excel or other data analysis programs, find a way to present the data to show that the size of the price shock, rather than the price level, is associated with the likelihood of revolution.
3. Do you think this is a plausible explanation for the revolutions that occurred?
4. A journalist suggests that similar factors played a part in the Arab Spring in 2010. Read the post. What do you think of this hypothesis?

		Average Price 1838–45	Maximum Price 1845–48
Violent revolution 1848	Austria	52.9	104.0
	Baden	77.0	136.6
	Bavaria	70.0	127.3
	Bohemia	61.5	101.2
	France	93.8	149.2
	Hamburg	67.1	108.7
	Hesse-Darmstadt	76.7	119.7
	Hungary	39.0	92.3
	Lombardy	88.3	119.1
	Mecklenburg-Schwerin	72.9	110.9
	Papal States	74.0	105.1
	Prussia	71.2	110.7
	Saxony	73.3	125.2
	Switzerland	87.9	146.7
	Württemberg	75.9	128.7
		Average Price 1838–45	**Maximum Price 1845–48**
Immediate constitutional change 1848	Belgium	93.8	140.1
	Bremen	76.1	109.5
	Brunswick	62.3	100.3
	Denmark	66.3	81.5
	Netherlands	82.6	136.0
	Oldenburg	52.1	79.3
		Average Price 1838–45	**Average Price 1845–48**
No revolution 1848	England	115.3	134.7
	Finland	73.6	73.7
	Norway	89.3	119.7
	Russia	50.7	44.1
	Spain	105.3	141.3
	Sweden	75.8	81.4

Figure 8.15 Average and peak prices of wheat in Europe, 1838–1845.

Helge Berger and Mark Spoerer. 2001. 'Economic Crises and the European Revolutions of 1848.' *The Journal of Economic History* 61 (2): pp. 293–326.

QUESTION 8.5 CHOOSE THE CORRECT ANSWER(S)

LINK

Look again at Figure 8.9, which shows the equilibrium of the bread r
loaves of bread per day at price Rs. 2. A year later, we find that the market equilibrium price has fallen to Rs. 1.50. What can we definitely conclude?

Continue from your last visit?

- [ ] The fall in the price must have been caused by a downward shift in the demand curve.
- [ ] The fall in the price must have been caused by a downward shift in the supply curve.
- [ ] The fall in price could have been caused by a shift in either curve.
- [ ] At a price of Rs. 1.50, there will be an excess demand for bread.

Check my answers

QUESTION 8.6 CHOOSE THE CORRECT ANSWER(S)

LINK

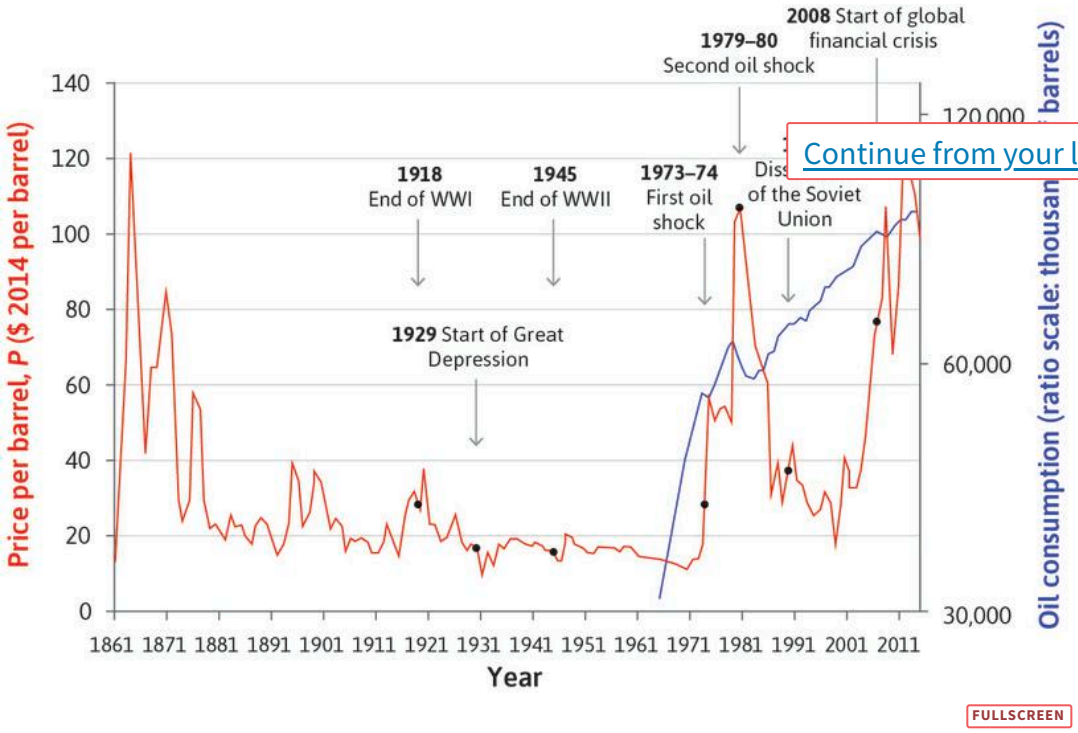
Which of the following statements are correct?

- [ ] A fall in the mortgage interest rate would shift up the demand curve for new houses.
- [ ] The launch of a new Sony smartphone would shift up the demand curve for existing iPhones.
- [ ] A fall in the oil price would shift up the demand curve for oil.
- [ ] A fall in the oil price would shift down the supply curve for plastics.

Check my answers

8.6 The world oil market

Figure 8.16 plots the real price of oil in world markets (in constant 2014 US dollars) and the total quantity consumed globally from 1865 to 2014. To understand what drives the large fluctuations in the oil price, we can use our supply and demand model, distinguishing between the short run and the long run.



Continue from your last visit?

Figure 8.16 World oil prices in constant prices (1865–2014) and global oil consumption (1965–2014).

BP. (2015) *BP Statistical Review of World Energy June 2015*.

Prices reflect **scarcity**. If a good becomes scarcer or costlier to produce, then the supply will fall and the price will tend to rise. For more than 60 years, oil industry analysts have been predicting that demand would soon outstrip supply: production would reach a peak and prices would then rise as world reserves declined. 'Peak oil' is not evident in Figure 8.16. One reason is that rising prices provide incentives for further exploration. Between 1981 and 2014, more than 1,000 billion barrels were extracted and consumed, yet world reserves of oil more than doubled from roughly 680 billion barrels to 1,700 billion barrels.

Prices have risen strongly in the twenty-first century and an increasing number of analysts are predicting that conventional oil, at least, has reached a peak. But unconventional resources such as shale oil are now being exploited. Perhaps it will be climate change policies, rather than resource depletion, that eventually curb oil consumption. [4] [5]

The oil price data in Figure 8.16 is difficult to interpret because of the sharp swings from high to low and back again over short periods of time. These fluctuations cannot be explained by looking at oil reserves, because they reflect short-run scarcity. Both supply and demand are inelastic in the short run.

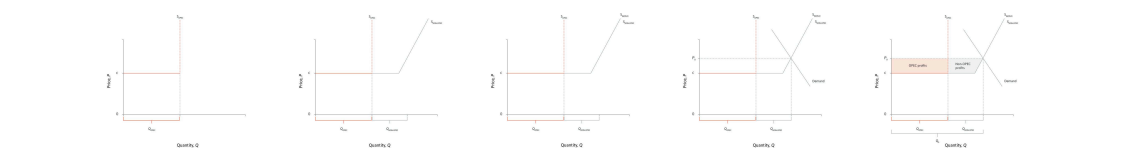
Short-run supply and demand LINK

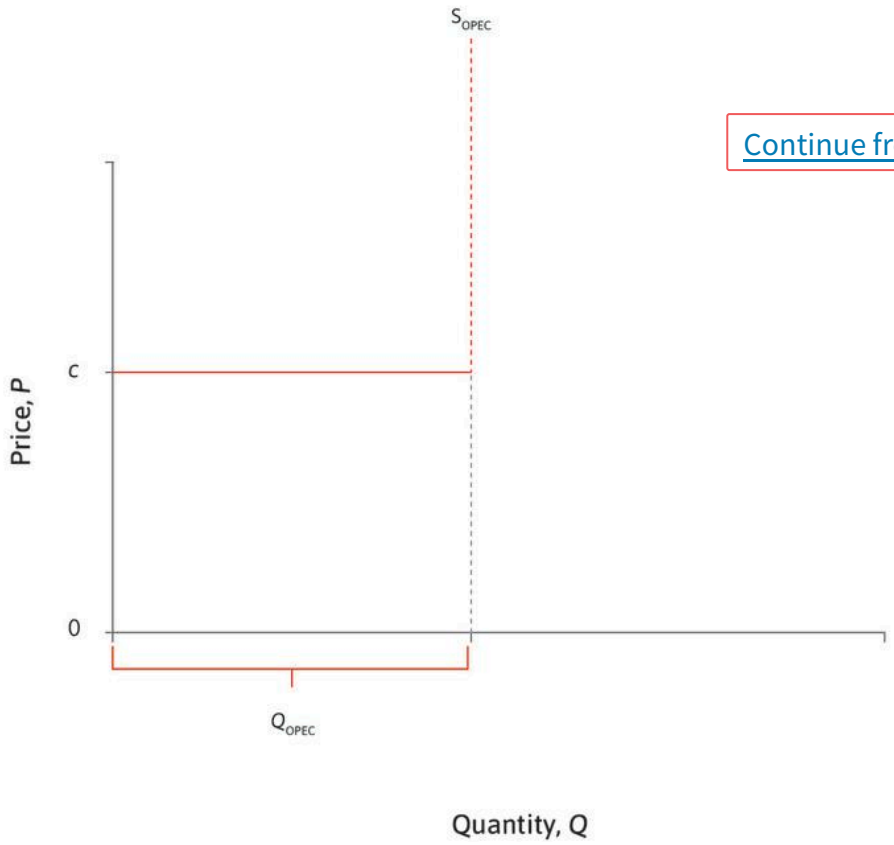
On the demand side, the main use of oil products is in transport services (air, road, and sea). Demand is inelastic in the short run because of the limited substitution possibilities. For example, even if petrol prices rise substantially, commuters will continue to use their existing cars to travel to work because of the limited alternatives immediately available to them. Therefore, the short-run demand curve is steep.

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Traditional oil extraction technology is characterized by a large up-front investment in expensive oil wells that can take many months or longer to construct, and once in place, can keep pumping until the well is depleted or oil can no longer be profitably extracted. Once the well is drilled, the cost of extracting the oil is relatively low, but the rate at which the oil is pumped faces capacity constraints—producers can get only so many barrels per day from a well. This means that, taking existing capacity as fixed in the short run, we should draw a short-run market supply curve that is initially low and flat, and then turns upwards very steeply as capacity constraints are hit. We also need to allow for the **oligopolistic** structure of the world market for crude oil. The Organization of Petroleum Exporting Countries (OPEC) is a cartel with a dozen member countries that currently accounts for about 40% of world oil production. OPEC sets output quotas for its members. We can represent this in our supply and demand diagram by a flat marginal cost line that stops at the total OPEC production quota. At that point, the line becomes vertical. This is not because of capacity constraints, but because OPEC producers will not sell any more oil.

Figure 8.17 assembles the market supply curve by adding the OPEC production quota to the supply from non-OPEC countries (remember that we obtain market supply curves by adding the amounts supplied by each producer at each price) and combines it with the demand curve to determine the world oil price.





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FULLSCREEN

OPEC supply

OPEC's members can increase production easily within their current capacity, without increasing their marginal cost c. OPEC quotas limit their total production to Q_{OPEC}.

Figure 8.17 The world market for oil.

The 1970s oil price shocks LINK

In 1973 and 1974, OPEC countries imposed a partial oil embargo in response to the Middle East war. In 1979 and 1980, oil production by Iran and Iraq fell because of the supply disruptions following the Iranian Revolution and the outbreak of the Iran–Iraq war. These are represented in Figure 8.18 by a leftward shift of the world supply curve S_{world}, driven by a reduction in the volume of OPEC production to Q'_{OPEC}. Total production and consumption falls, but because demand is very price inelastic, the percentage increase in price is much larger than the percentage decrease in quantity. This is what we see in the data in Figure 8.18. The oil price (in 2014 US dollars) goes from $18 per barrel in 1973 to $56 in 1974, and then to $106 in 1980, but the declines in world oil consumption after these price shocks are small by comparison (–2% between 1973 and 1975, and –10% between 1979 and 1983).

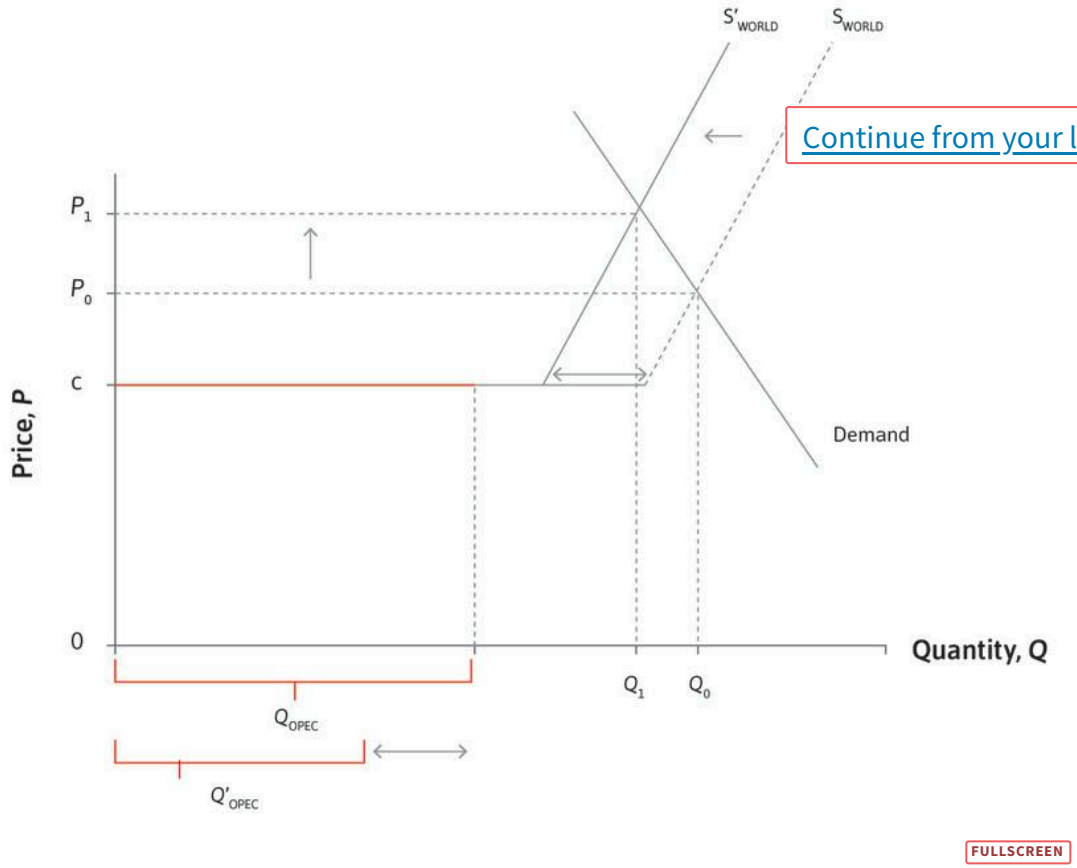


Figure 8.18 The OPEC oil price shocks of the 1970s: OPEC decreases output.

The 2000–2008 oil price shocks LINK

The years 2000 to 2008 were a period of rapid economic growth in industrializing countries, especially China and India. The **income elasticity of demand** for oil and oil products is higher in these countries than in developed market economies, and demand for car ownership and tourist air travel is growing relatively rapidly as the countries become wealthier. This increase in income moves the demand curve to the right, as shown in Figure 8.18. In this case, it is the inelastic short-run supply curve for oil that accounts for the big increase in price and only a modest increase in world oil consumption. The sharp price decrease in 2009 has the same explanation, but in reverse —the financial crisis of 2008–9 was a negative demand shock that moved the demand curve to the left, so world consumption fell by about 3%, and the price of crude fell from over $100 per barrel in the summer of 2008 to $40–50 in early 2009.

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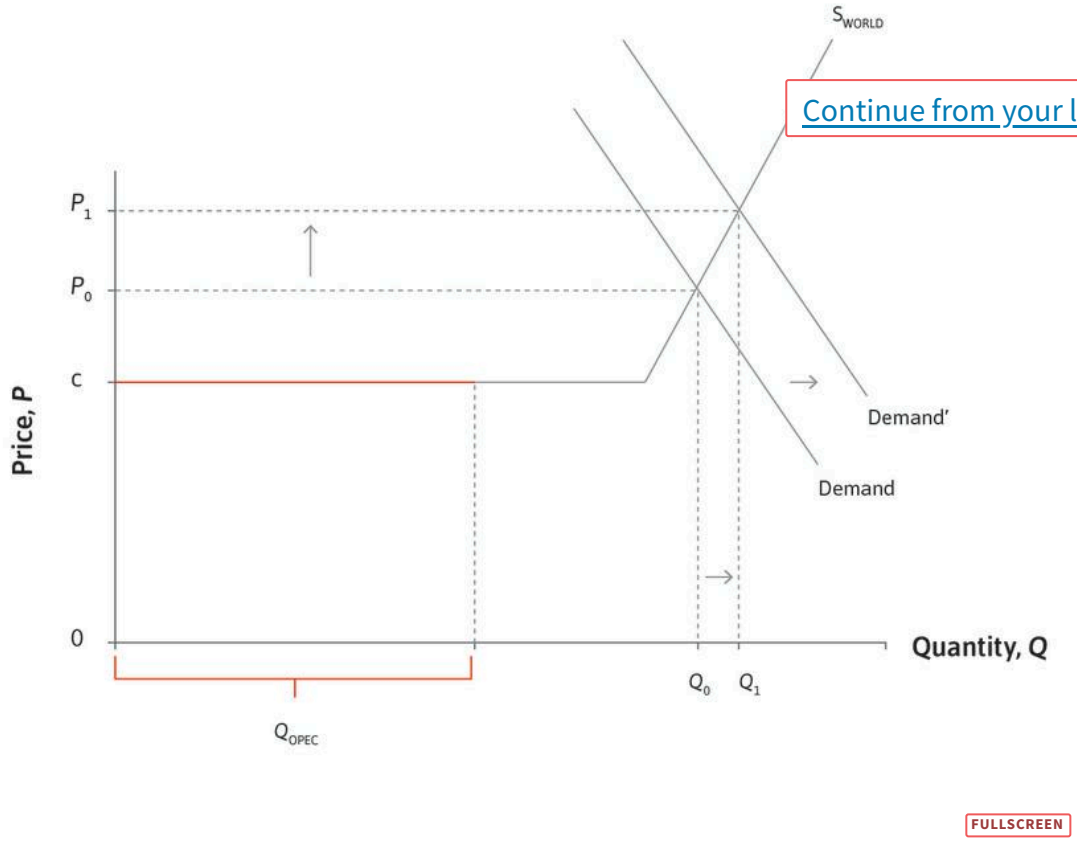


Figure 8.19 The oil price shocks of 2000–8: Economic growth increases world demand.

EXERCISE 8.6 THE WORLD MARKET FOR OIL LINK

Using a supply and demand diagram:

1. Illustrate what happens when economic growth boosts world demand
   a. in the short run
   b. in the long run as producers invest in new oil wells
   c. in the long run as consumers find substitutes for oil
2. Similarly, describe the short- and long-run consequences of a negative supply shock similar to the 1970s shock.
3. If you observed an oil price rise, how in principle could you tell whether it was driven by supply-side or demand-side developments?
4. How would the diagram, and the response to shocks, be different if there were:
   a. a competitive market composed of many producers?
   b. a single monopoly oil producer?
   c. an OPEC cartel controlling 100% of world oil production and seeking to maximize the combined profits of its members?
5. Why would individual OPEC member countries have an incentive to produce more than the quota assigned to them?
6. Does this logic carry over to the situation in the real world where there are also non-OPEC producers?

EXERCISE 8.7 THE SHALE OIL REVOLUTION LINK

An important development in the past 10 years has been the re-emergence of the US as a major oil producer via the 'shale oil revolution'. Shale oil is extracted [illegible] of hydraulic fracturing or 'fracking'—injecting fluid into ground at hi[illegible] the rock and allow extraction. In a speech called 'The New Economics of Oil' in October 2015, Spencer Dale, group chief economist at oil producer BP PLC, explained how shale oil production differs from traditional extraction.

1. According to Dale, how has the shale oil revolution affected the world market for oil?
2. How will the world oil market be different in future?
3. Explain how our supply and demand diagram should be changed if his analysis is correct.

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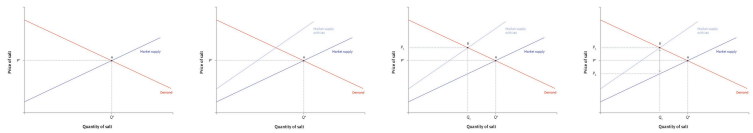
8.7 The effects of taxes

Governments can use taxation to raise revenue (to finance government spending, or redistribute resources) or to affect the allocation of goods and services in other ways, perhaps because the government considers a particular good to be harmful. The supply and demand model is a useful tool for analysing the effects of taxation.

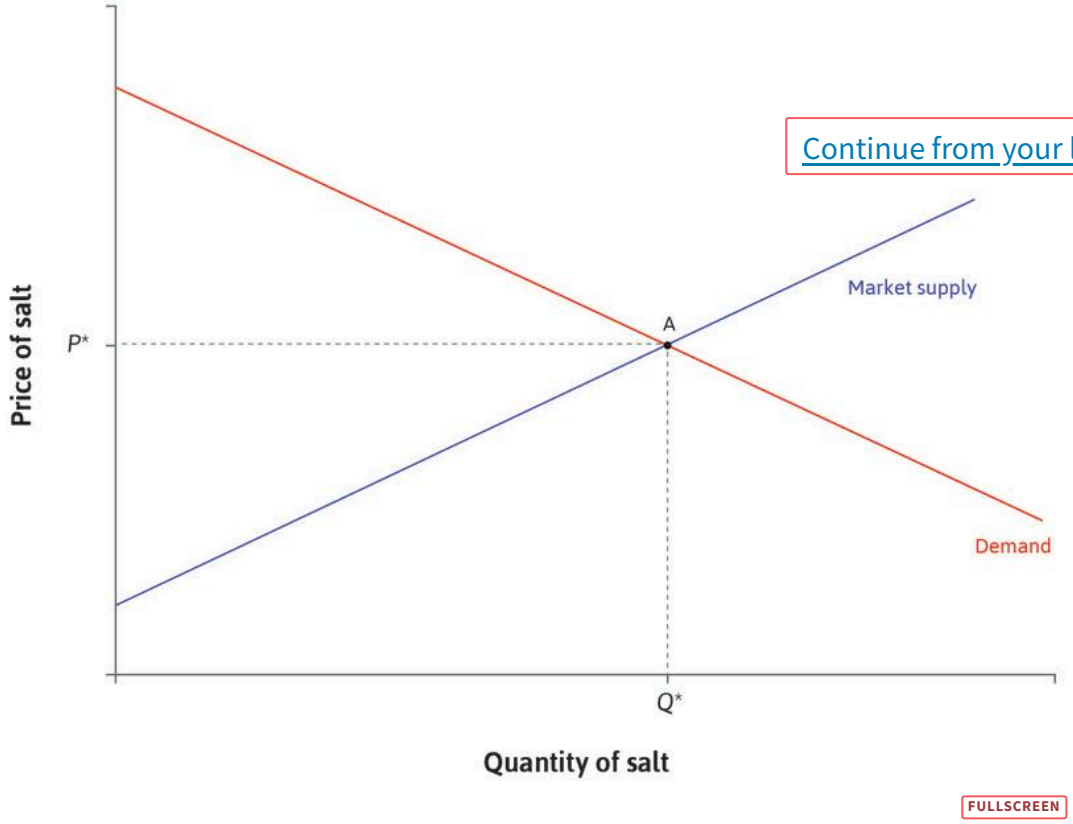
Using taxes to raise revenue LINK

Raising revenue through taxation has a long history (see Unit 22 of *The Economy*). Take the taxation of salt, for example. For most of history, salt was used all over the world as a preservative, allowing food to be stored, transported, and traded. The ancient Chinese advocated taxing salt, since people needed it, however high the price. Salt taxes were an effective but often resented tool used by ruling elites in ancient India and medieval kings. Resentment of high salt taxes played an important part in the French Revolution, and Gandhi led protests against the salt tax imposed by the British in India.

Figure 8.20 illustrates how a salt tax might work. Initially the market equilibrium is at point A: the price is P^* and the quantity of salt traded is Q^*. Suppose that a sales tax of 30% is imposed on the price of salt, to be paid to the government by the suppliers. If suppliers have to pay a 30% tax, their marginal cost of supplying each unit of salt increases by 30%. So the supply curve shifts: the price is 30% higher at each quantity.



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The initial equilibrium

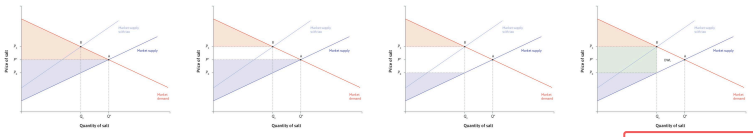
Initially the market equilibrium is at point A. The price is P^* and the quantity of salt sold is Q^*.

Figure 8.20 The effect of a 30% salt tax.

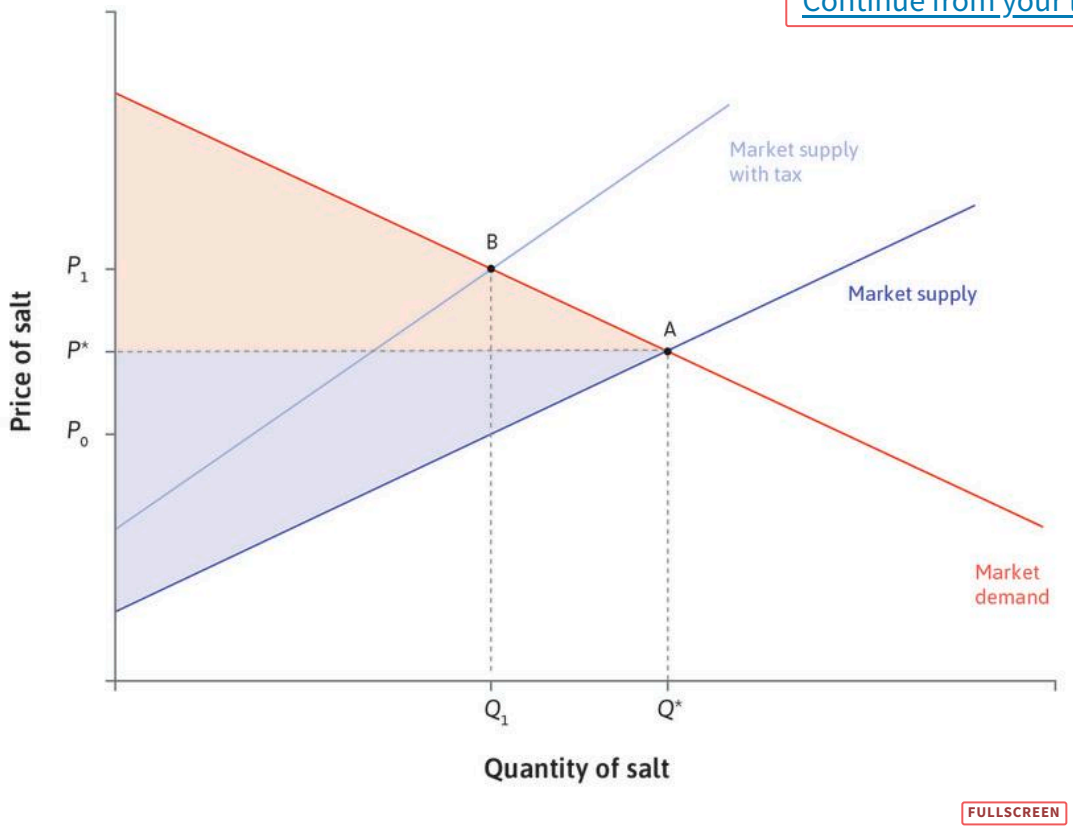
The new equilibrium is at point B, where a lower quantity of salt is traded. Although the consumer price has risen, note that it is not 30% higher than before. The price paid by consumers, P_1, is 30% higher than the price received by the suppliers (net of the tax), which is P_0. Suppliers receive a lower price than before, they produce less, and their profits will be lower. This illustrates an important feature of taxes: it is not necessarily the taxpayer who feels its main effect. In this case, although the suppliers pay the tax, the **tax incidence** falls partly on consumers and partly on producers.

Figure 8.21 shows the effect of the tax on consumer and producer surplus:

- *Consumer surplus falls*: Consumers pay a higher price, and buy less salt.
- *Producer surplus falls*: They produce less and receive a lower net price.
- *Total surplus is lower*: Even taking account of the tax revenue received by the government, the tax causes a deadweight loss.



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Maximized gains from trade

Before the tax is imposed, the equilibrium allocation at A maximizes the gains from trade. In the upper panel the red triangle is the consumer surplus and the blue triangle is the producer surplus.

Figure 8.21 Taxation and deadweight loss.

When the salt tax is imposed, the total surplus from trade in the salt market is given by:

$$\text{total surplus} = \text{consumer surplus} + \text{producer surplus} + \text{government revenue}$$

Since the quantity of salt traded is no longer at the level that maximizes gains from trade, the tax has led to a deadweight loss.

In general, taxes change prices, and prices change buyers' and sellers' decisions, which can cause deadweight loss. To raise as much revenue as possible, the government would prefer to tax a good for which demand is not very responsive to price, so that the fall in quantity traded is quite small—that is to say, a good with a low elasticity of demand. That is why the ancient Chinese recommended taxing salt.

We can think of the total surplus as a measure of the welfare of society as a whole (provided that the tax revenue is used for the benefit of society). So there is a second reason for a government that cares about welfare to prefer taxing goods with low elasticity of demand—the loss of total surplus will be lower. The

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depends on what the government does with the revenues that it collects:

- *The government spends the revenue on goods and services that enhance the wellbeing of the population*: Then the tax and resulting expenditure may enhance public welfare—even though it reduces the surplus in the particular market that is taxed.
- *The government spends the revenues on an activity that does not contribute to wellbeing*: Then the lost consumer surplus is just a reduction in the living standards of the population.

Therefore, taxes can improve or reduce overall welfare. The most that we can say is that taxing a good whose demand is inelastic is an efficient way to transfer the surplus from consumers to the government.

The government's power to levy taxes is a bit like the price-setting power of a firm that sells a differentiated good. It uses its power to raise the price and collect revenue, while reducing the quantity sold. Its ability to levy taxes depends on the institutions it can use to enforce and collect them.

One reason for the use of salt taxes in earlier times was that it was relatively easy for a powerful ruler to take full control of salt production, in some cases as a monopolist. In the notorious case of the French salt tax, the monarchy not only controlled all salt production; it also forced its subjects to buy up to 7 kg of salt each per year.

In March and April 1930, the artificially high price of salt in British colonial India provoked one of the defining moments of the Indian independence movement: Mahatma Gandhi's salt march to acquire salt from the Indian ocean. Similarly, in what came to be called the Boston tea party, in 1773 American colonists objecting to a British colonial tax on tea dumped a cargo of tea into the Boston harbour.

Resistance to taxes on inelastic goods arises for the very reason they are imposed: they are difficult to escape!

In many modern economies the institutions for tax collection are well-established, usually with democratic consent. Provided that citizens believe taxes have been implemented fairly, using them to raise revenue is accepted as a necessary part of social and economic policy. We will now look at another reason why governments may decide to levy taxes.

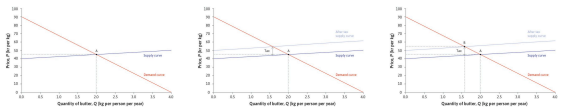
Using taxes to change behaviour LINK

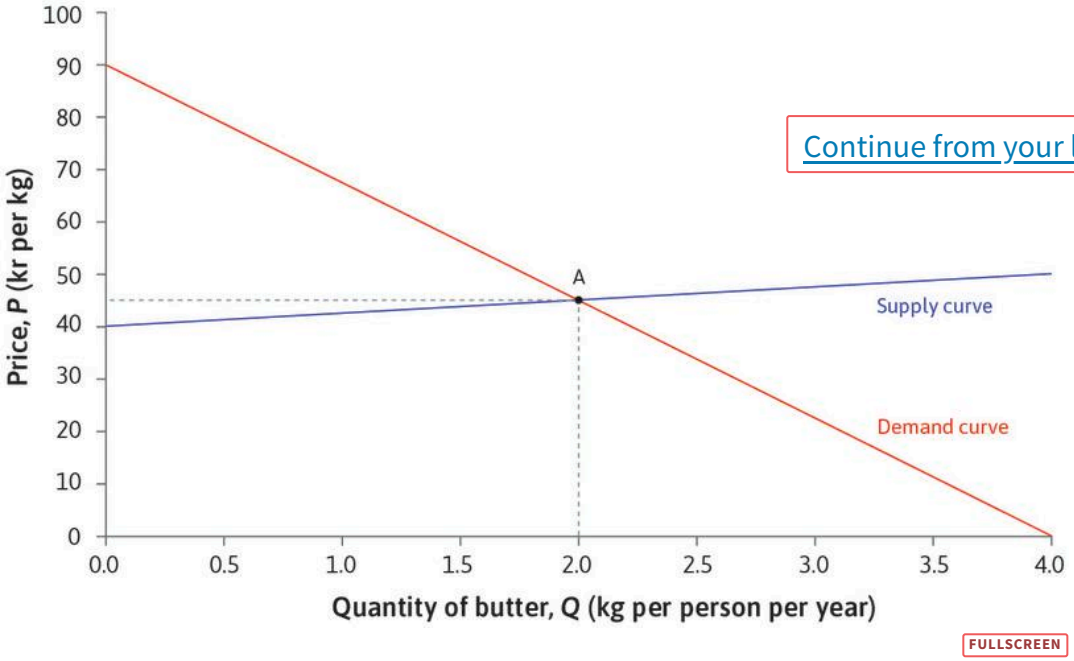
Policymakers in many countries are interested in the idea of using taxes to deter consumption of unhealthy foods with the objective of improving public health and tackling the obesity epidemic. Some countries have introduced f
including France, Norway, Mexico, Samoa, and Fiji, tax sweetened drinks. Hungary's 'chips tax' is aimed at products carrying proven health risks, particularly those with high sugar or salt content. In 2011, the Danish government introduced a tax on products with high saturated fat content. 6

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The level of the Danish tax was 16 Danish kroner (kr) per kilogram of saturated fat, corresponding to 10.4 kr per kg of butter. Note that this was a *specific tax*, levied as a fixed amount per unit of butter. A tax like the one we analysed for salt, levied as a percentage of the price, is known as an *ad valorem tax*. According to a study of the Danish fat tax, it corresponded to about 22% of the average butter price in the year before the tax. The study found that it reduced the consumption of butter and related products (butter blends, margarine, and oil) by between 15% and 20%. We can illustrate the effects in the same way as we did for the salt tax, using the supply and demand model (we are assuming here that butter retailers are price-takers).

Figure 8.22 shows a demand curve for butter, measured in kilograms per person per year. The numbers correspond approximately to Denmark's experience. We have drawn the supply curve for butter as almost flat, on the assumption that the marginal cost of butter for retailers does not change very much as quantity varies. The initial equilibrium is at point A, where the price of butter is 45 kr per kg, and each person consumes 2 kg of butter per year.





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Equilibrium in the market for butter

Initially the market for butter is in equilibrium. The price of butter is 45 kr per kg, and consumption of butter in Denmark is 2 kg per person per year.

Figure 8.22 The effect of a fat tax on the retail market for butter.

A tax of 10 kr per kg shifts the supply curve upwards and leads to a rise in price to 54 kr, and a fall in consumption to 1.6 kg. The consumer price rises by 9 kr—almost the full amount of the tax—and the suppliers' net revenue per kg of butter falls to 44 kr. In this case, although suppliers pay the tax, the tax incidence is felt mainly by consumers. Of the 10 kr tax per kg, the consumer effectively pays 9 kr, while the supplier or producer pays 1 kr. So the price received by the retailers, net of tax, is only 1 kr lower.

Figure 8.23 shows what happens to consumer and producer surplus as a result of the fat tax.

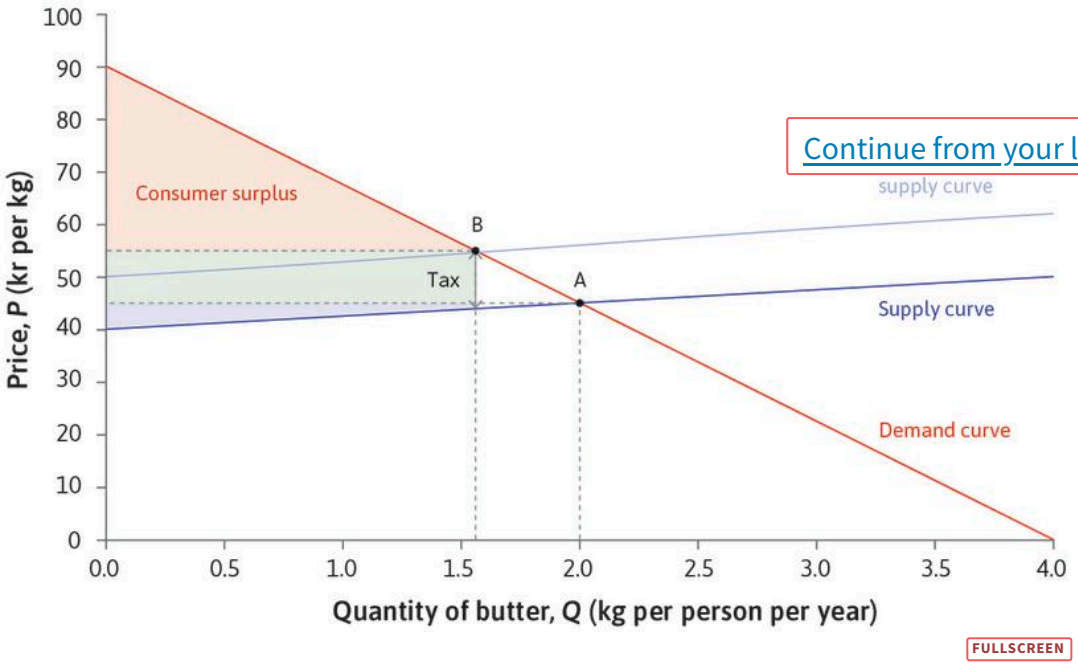


Figure 8.23 The effect of a fat tax on the consumer and producer surplus for butter.

Again, both consumer and producer surpluses fall. The area of the green rectangle represents the tax revenue: with a tax of 10 kr per kg and equilibrium sales of 1.6 kg per person, tax revenue is 10 × 1.6 = 16 kr per person per year.

How effective was the fat tax policy? For a full evaluation of the effect on health we should look at all the foods taxed, and take into account the cross-price effects—the changes in consumption of other foods caused by the tax. The study of the Danish tax also allowed for the possibility that some retailers are not price-takers. Nevertheless, Figures 8.16 and 8.17 illustrate some important implications of the tax:

- *Consumption of butter products fell*: In this case by 20%. You can see this in Figure 8.16. In this respect, the policy was successful.
- *There was a large fall in surplus, especially consumer surplus*: You can see this in Figure 8.17. But recall that the government's aim when it implemented the fat tax policy was not to raise revenue, but rather to reduce quantity. So the fall in consumer surplus was inevitable. The loss of surplus caused by a tax is a deadweight loss, which sounds negative. But in this case the policymaker might see it as a gain if the 'good', butter, is considered 'bad' for consumers.

One aspect of taxation not illustrated in our supply and demand analysis is the cost of collecting it. Although the Danish fat tax successfully reduced fat consumption, the government abolished it after only 15 months because of the administrative burden it placed on firms. Any taxation system requires effective mechanisms for tax collection, and designing taxes that are simple to administer (and difficult to avoid) is an important goal of tax policy. Policymakers who want to introduce food taxes will need

to find ways of minimizing administrative costs. But since the costs cannot be eliminated, they will also need to consider whether the health gain (and reduction of costs of bad health) will be sufficient to offset them.

Continue from your last visit?

EXERCISE 8.8 THE DEADWEIGHT LOSS OF THE BUTTER TAX

LINK

Food taxes such as the ones discussed here and in Unit 7 are often intended to shift consumption towards a healthier diet, but give rise to deadweight loss.

Why do you think a policymaker and a consumer might interpret this deadweight loss differently?

QUESTION 8.7 CHOOSE THE CORRECT ANSWER(S)

LINK

Figure 8.20 shows the demand and supply curves for salt, and the shift in the supply curve due to the implementation of a 30% tax on the price of salt. Which of the following statements are correct?

- [ ] In the post-tax equilibrium, the consumers pay P_1 and the producers receive P^*.
- [ ] The government's tax revenue is given by $(P^* - P_0)Q_1$.
- [ ] The deadweight loss is given by $(1/2)(P_1 - P_0)(Q^* - Q_1)$.
- [ ] As a result of the tax, the consumer surplus is reduced by $(1/2)(Q_1 + Q^*)(P_1 - P^*)$.

Check my answers

QUESTION 8.8 CHOOSE THE CORRECT ANSWER(S)

LINK

Figure 8.23 shows the effect of a tax intended to reduce the consum

Continue from your last visit?

before-tax equilibrium is at A = (2.0 kg, 45 kr) and the after-tax equilibrium is at B = (1.6 kg, 54 kr). The tax imposed is 10 kr per kg of butter. Which of the following statements is correct?

- [ ] The producers receive 45 kr per kg of butter.
- [ ] The tax policy would be more effective if the supply curve were less elastic.
- [ ] The very elastic supply curve implies that the incidence of the tax falls mainly on consumers.
- [ ] The loss of consumer surplus due to tax is (1/2) × 10 × (2.0 – 1.6) = 2.0.

Check my answers

8.8 Price-setting and price-taking firms

We now have two different models of how firms behave. In the Unit 7 model, the firm produces a product that is different from the products of other firms, giving it market power—the power to set its own price. This model applies to the extreme case of a monopolist, who has no competitors at all, such as water supply companies, and national airlines with exclusive rights granted by the government to operate domestic flights. The Unit 7 model also applies to a firm producing differentiated products such as Cheerios or language courses—similar, but not identical, to those of its competitors. In such cases, the firm still has the power to set its own price. But if it has close competitors, demand will be quite elastic and the range of feasible prices will be narrow.

In the supply-and-demand model developed in this unit, firms are price-takers. Competition from other firms producing identical products means that they have no power to set their own prices. This model can be useful as an approximate description of a market in which there are many firms selling very similar products, even if the idealized conditions for a perfectly competitive market do not hold.

In practice, economies are a mixture of more and less competitive markets. In some respects, firms act the same whether they are the single seller of a good or one of a great many competitors: all firms decide how much to produce, which technologies to use, how many people to hire, and how much to pay them so as to maximize their profits. Joan Robinson pioneered the economic theory of market competition among firms that were neither monopolies nor the price-taking firms that are the basis of the model of perfect competition that we studied in this unit.

But there are important differences. Look back at the decisions made by price-setting firms to maximize profits (Figure 7.2). Firms in more competitive markets lack either the incentive or the opportunity to do some of these things.

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A firm with a unique product will advertise (Buy Nike!) to shift the demand curve for its product to the right. But why would a single competitive firm advertise (Drink milk!)? This would shift the demand curve for all of the firms in the industry. Advertising in a competitive market is a **public good**: the benefits go to all of the firms in the industry. If you see a message like 'Drink milk!' it is probably paid for by an association of dairies, not by a particular one.

The same is true of expenditures to influence public policy. If a large firm with market power is successful, for example, in relaxing environmental regulations, then it will benefit directly. But activities like lobbying or contributing money to electoral campaigns will be unattractive to the competitive firm because the result (a more profit-friendly policy) is a public good.

Similarly, investment in developing new technologies is likely to be undertaken by firms facing little competition, because if they are successful in finding a profitable innovation, the benefits will not be lost to competitors also adopting it. However, one way that successful large firms can emerge is by breaking away from the competition and innovating with a new product. The UK's largest organic dairy, Yeo Valley, was once an ordinary farm selling milk, just like thousands of others. In 1994 it established an organic brand, creating new products for which it could charge premium prices. With the help of imaginative marketing campaigns it has grown into a company with 1,800 employees and 65% of the UK organic market.

The table in Figure 8.24 summarizes the differences between price-setting and price-taking firms.

Price-setting firm or monopoly	Firm in a perfectly competitive market
Sets price and quantity to maximize profits ('price-maker')	Takes market determined price as given and chooses quantity to maximize profits ('price-taker')
Chooses an output level at which marginal cost is less than price	Chooses an output level at which marginal cost equals price
Deadweight losses (Pareto inefficient)	No deadweight losses for consumers and firms (can be Pareto efficient if no one else in the economy is affected)
Owners receive economic rents (profits greater than normal profits)	If the owners receive economic rents, the rents are likely to disappear as more firms enter the market
Firms advertise their unique product	Little advertising: it costs the firm, but benefits all firms (it's a public good)
Firms may spend money to influence elections, legislation and regulation	Little expenditure by individual firms on this (same as advertising)
Firms invest in research and innovation; seek to prevent copying	Little incentive for innovation; others will copy (unless the firm can succeed in differentiating its product and escaping from the competitive market)

Continue from your last visit?

Figure 8.24 Price-setting and price-taking firms.

GREAT ECONOMISTS

Joan Robinson LINK

A letter to a female student in 1970 from Paul Samuelson, perhaps the most influential economist of the twentieth century, concluded: 'P.S. Do study economics. Perhaps the best economist in the world happens also to be a woman (Joan Robinson).'

That Robinson's much lauded intellectual achievements were not crowned with a Nobel prize has drawn many speculations. Was it because she was a woman in a male-dominated discipline? Because of her late endorsement of the Maoist cultural

revolution? Or because of her relentless critique of what she called 'mainstream' economics (including, very pointedly, Samuelson's ideas)?



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Her caustic remarks about the economics profession won her both followers and detractors. She wrote: 'For many years I have been employed as a teacher of theoretical economics; I would like to believe that I earn my living honestly, but I often have doubts'. [7] She also wrote about economists: 'The purpose of studying economics is not to acquire a set of ready-made answers to economic questions, but to learn how to avoid being deceived by economists'. [8]

Her many theoretical contributions threw her into more than one controversy. As a member of the small circle at the University of Cambridge whom John Maynard Keynes drew upon to comment on and refine *General Theory*, she made his work accessible in her 1937 book *Introduction to the Theory of Employment*. She quickly set out to extend Keynes' static theory in the long run, which entailed developing a dynamic theory of capital accumulation.

The publication of *The Accumulation of Capital* and related articles in the 1950s ignited a decade long debate with Samuelson and a colleague at MIT, Robert Solow, known as the 'Capital Controversy'. Robinson pointed to some circularity in how their 'quantity of capital' variable was defined. Since there were many different types of capital in the economy, the variable was calculated through aggregating their prices, which was then used to determine the return on capital, hence its price. Samuelson and Solow conceded that the problem in their theory of production and distribution was real, but also minor, and did not alter their growth model.

From her first major publication, perceptively entitled 'Economics is a Serious Subject' (1932) to her late series of publications on *Economic Philosophy* (1964), *Economics: An Awkward Corner* (1966), and *What are the Questions?* (1977), Robinson relentlessly insisted economists should take into account time, history, and institutions and consider the welfare and political implications of economic analysis. In later years, she criticized her earlier work on imperfect competition for being too 'scholastic', 'primitive', and not 'radical' enough.

Her advice to teachers of economics was to 'start from the beginning to discuss various types of economic system. Every society (except Robinson Crusoe) has to have some rules of the game for organizing production and the distribution of the product… [and] displace the theory of the relative prices of commodities from the

centre of the picture...' [9] The order of topics in the CORE introduction to economics follows this logic.

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8.9 Prices and quantities with an increase in demand

Earlier in the unit, we examined what happened to the price of quinoa when there was an increase in demand in the market for quinoa—that is, when demand was higher at each possible price. We saw that the new equilibrium was at a higher price and quantity.

This occurred because to produce more quinoa farmers had to cultivate less suitable land so the marginal cost of producing the crop rose. The shape of the supply curve reflected the marginal cost of production. As the demand curve shifted, the market price rose along the rising supply curve.

We often do observe price rises when demand increases, but this doesn't necessarily happen. It is possible that in certain cases prices could stay the same, or even fall. Why might this happen?

To understand how prices respond to demand, we need to think about how costs change as quantity rises. We can see the range of possible outcomes if we consider a price-setting firm, like Language Perfection (LP) in Unit 7.

Figure 8.25 shows the initial profit-maximising choice of price and quantity for LP at point A, where the the original demand curve is tangent to an isoprofit curve. What will happen if the demand curve pivots outwards, so that demand is higher at each price? That depends on the shape of LP's isoprofit curves, and the shape of the isoprofit curves depends on LP's marginal costs.

We assumed in Unit 7 that LP could produce language courses at a constant marginal cost—that is, the cost per course does not change as the number of courses offered increases. But there are other possibilities. The marginal cost of a course might increase if LP has to raise wages to attract more tutors. In this case, the isoprofit curves will begin to slope upwards at higher numbers of students, because with higher marginal cost the price has to rise to achieve the same level of profit. Point B shows the profit-maximising choice when the isoprofit curves have this shape; you can see that the price is higher than it was initially at point A. So this case is similar to the quinoa example.

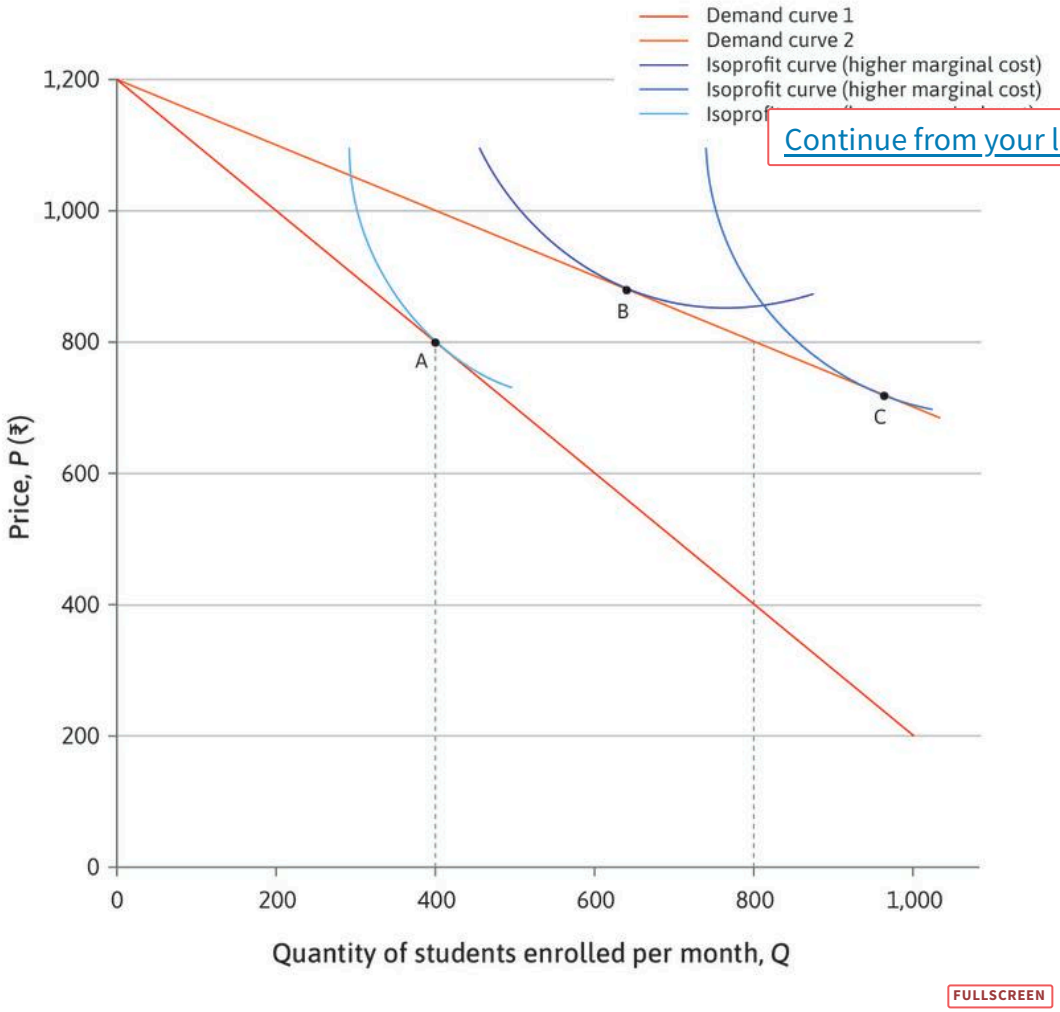


Figure 8.25 Increases in demand have ambiguous effects on price.

Alternatively, marginal costs might decrease with quantity. For example, if larger scale production of course materials can be done at lower cost per student. Then, the isoprofit curves will tend to slope down more steeply at higher quantities. In this case the increase in demand would lead to a profit-maximising choice of point C, where the price is lower than it was initially.

It can also be shown that, if LP has constant marginal costs as in Unit 7, a demand increase like the one shown in Figure 8.25 will lead to price staying exactly the same as before, while the quantity of courses increases.

This example illustrates that if we want to predict the effect on a market price of increased demand, we should first think about how production costs change with the total quantity produced. If you want to know more about how firms' cost change with output, and how this affects the shape of their isoprofit curves, you can read about the example of a firm called Beautiful Cars, in sections 7.3 and 7.4 of *The Economy*, and watch this video.

8.10 Conclusion

To our economic toolkit we have added two models of firm behaviour, each relying on different assumptions regarding the nature of the product and t

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	Price-setting firm (monopoly)	Price-taking firm
Setting and assumptions	The firm has few competitors as it produces a differentiated product. It faces a downward-sloping product demand curve. The firm sets price to maximize profits (price-setter). There is no price discrimination, so the chosen price is the same for all customers.	Competition from other firms producing identical products means that firms have no power to set their own prices. They each face a flat demand curve for their product. Given the market price, the firm chooses the quantity to produce to maximize profits (price-taker).
Economic toolkit	**Constrained optimization problem** The firm chooses the highest isoprofit curve possible, given the product demand curve as a constraint. The profit-maximizing choice is where the curves are tangent. This is where MRS (slope of isoprofit) equals MRT (slope of demand).	**Supply and demand analysis** The supply curve depends on suppliers' willingness to accept (their reservation prices). Its shape depends on the marginal cost curve. The market-clearing price is determined by the intersection of market supply and market demand curves.
Main result	Price is greater than marginal cost, and owners receive economic rents. There exist deadweight losses, meaning there are unexploited gains from trade. The outcome is Pareto inefficient.	Price is equal to marginal cost. In this competitive equilibrium, total surplus is maximized and the outcome is Pareto efficient, assuming only buyers and sellers are affected. Owners can only receive dynamic rents when markets are in disequilibrium following an exogenous shock.

A firm's market power determines the markup it can set, which is inversely related to the price elasticity of demand. Competition policy aims to prevent abuses of market power that may result from the formation of cartels. The model of perfect competition provides a useful benchmark against which to evaluate economi

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Concepts introduced in Unit 8 LINK

Before you move on, review these definitions:

- Price-taking firms
- Competitive equilibrium
- Exogenous shocks
- Model of perfect competition
- Taxation

8.11 References

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